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Square Roots and Cube Roots

IMPORTANT FACTS AND FORMULAE

I. Square Root: If $x^2 = y$, we say that the square root of y is x and we write, $\sqrt{y} = x$.

Thus, $\sqrt{4} = 2$, $\sqrt{9} = 3$, $\sqrt{196} = 14$.

II. Cube Root: The cube root of a given number x is the number whose cube is x . We denote the cube root of x by $\sqrt[3]{x}$.

Thus, $\sqrt[3]{8} = \sqrt[3]{2 \times 2 \times 2} = 2$, $\sqrt[3]{343} = \sqrt[3]{7 \times 7 \times 7} = 7$ etc.

Note:

$$1. \sqrt{xy} = \sqrt{x} \times \sqrt{y}$$

$$2. \sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}} = \frac{\sqrt{x}}{\sqrt{y}} \times \frac{\sqrt{y}}{\sqrt{y}} = \frac{\sqrt{xy}}{y}$$

SOLVED EXAMPLES

Ex. 1. Evaluate $\sqrt{6084}$ by factorization method.

Sol. Method: Express the given number as the product of prime factors. Now, take the product of these prime factors choosing one out of every pair of the same primes. This product gives the square root of the given number.

Thus, resolving 6084 into prime factors, we get:

$$6084 = 2^2 \times 3^2 \times 13^2$$

$$\therefore \sqrt{6084} = (2 \times 3 \times 13) = 78.$$

Ex. 2. Find the square root of 1471369.

Sol. Method: In the given number, mark off the digits in pairs starting from the unit's digit. Each pair and the remaining one digit is called a period.

Now, $1^2 = 1$. On subtracting, we get 0 as remainder.

Now, bring down the next period i.e., 47.

Now, trial divisor is $1 \times 2 = 2$ and trial dividend is 47.

So, we take 22 as divisor and put 2 as quotient.

The remainder is 3.

Next, we bring down the next period which is 13.

Now, trial divisor is $12 \times 2 = 24$ and trial dividend is 313. So, we take 241 as dividend and 1 as quotient.

The remainder is 72.

Bring down the next period i.e., 69.

Now, the trial divisor is $121 \times 2 = 242$ and the trial dividend is 7269. So, we take 3 as quotient and 2423 as divisor. The remainder is then zero.

$$\text{Hence, } \sqrt{1471369} = 1213.$$

Ex. 3. Evaluate: $\sqrt{248 + \sqrt{52 + \sqrt{144}}}$.

(P.C.S., 2009)

Sol. Given expression = $\sqrt{248 + \sqrt{52 + 12}} = \sqrt{248 + \sqrt{64}} = \sqrt{248 + 8} = \sqrt{256} = 16.$

2	6084
2	3042
3	1521
3	507
13	169
	13

1	1471369 (1213
1	
22	47
	44
241	313
	241
2423	7269
	7269
	x

Ex. 4. Simplify : $\frac{112}{\sqrt{196}} \times \frac{\sqrt{576}}{12} \times \frac{\sqrt{256}}{8}$.

Sol. Given expression = $\frac{112}{14} \times \frac{24}{12} \times \frac{16}{8} = 8 \times 2 \times 2 = 32$.

Ex. 5. If $a * b * c = \frac{\sqrt{(a+2)(b+3)}}{c+1}$, then find the value of $6 * 15 * 3$.

Sol. $6 * 15 * 3 = \frac{\sqrt{(6+2)(15+3)}}{3+1} = \frac{\sqrt{8 \times 18}}{4} = \frac{\sqrt{144}}{4} = \frac{12}{4} = 3$.

Ex. 6. Find the value of $\sqrt{1\frac{9}{16}}$.

Sol. $\sqrt{1\frac{9}{16}} = \sqrt{\frac{25}{16}} = \frac{\sqrt{25}}{\sqrt{16}} = \frac{5}{4} = 1\frac{1}{4}$.

Ex. 7. What is the square root of 0.0009?

Sol. $\sqrt{0.0009} = \sqrt{\frac{9}{10000}} = \frac{\sqrt{9}}{\sqrt{10000}} = \frac{3}{100} = 0.03$.

Ex. 8. Evaluate $\sqrt{175.2976}$.

Sol. Method: We make even number of decimal places by affixing a zero, if necessary. Now, we mark off periods and extract the square root as shown.

$\therefore \sqrt{175.2976} = 13.24$.

Ex. 9. What will come in place of question mark in each of the following questions?

(i) $\sqrt{\frac{32.4}{?}} = 2$ (ii) $\sqrt{86.49} + \sqrt{5 + (?)^2} = 12.3$

Sol. (i) Let $\sqrt{\frac{32.4}{x}} = 2$. Then, $\frac{32.4}{x} = 4 \Leftrightarrow 4x = 32.4 \Leftrightarrow x = 8.1$.

(ii) Let $\sqrt{86.49} + \sqrt{5 + x^2} = 12.3$.

Then, $9.3 + \sqrt{5 + x^2} = 12.3 \Leftrightarrow \sqrt{5 + x^2} = 12.3 - 9.3 = 3$

$\Leftrightarrow 5 + x^2 = 9 \Leftrightarrow x^2 = 9 - 5 = 4 \Leftrightarrow x = \sqrt{4} = 2$.

Ex. 10. Find the value of $\sqrt{\frac{0.289}{0.00121}}$.

(IGNOU, 2003)

Sol. $\sqrt{\frac{0.289}{0.00121}} = \sqrt{\frac{0.28900}{0.00121}} = \sqrt{\frac{28900}{121}} = \frac{170}{11}$.

Ex. 11. If $\sqrt{841} = 29$, then find the value of $\sqrt{841} + \sqrt{8.41} + \sqrt{0.0841} + \sqrt{0.000841}$.

(M.B.A., 2006)

Sol. Given expression = $\sqrt{841} + \sqrt{\frac{841}{10^2}} + \sqrt{\frac{841}{10^4}} + \sqrt{\frac{841}{10^6}} = \sqrt{841} + \frac{\sqrt{841}}{10} + \frac{\sqrt{841}}{10^2} + \frac{\sqrt{841}}{10^3}$
 $= 29 + \frac{29}{10} + \frac{29}{100} + \frac{29}{1000} = 29 + 2.9 + 0.29 + 0.029 = 32.219$.

Ex. 12. If $\sqrt{1 + \frac{x}{144}} = \frac{13}{12}$, then find the value of x .

1	$\overline{175.2976}$ (13.24)
	1
23	$\overline{75}$
	69
262	$\overline{629}$
	524
2644	$\overline{10576}$
	10576
	$\times$

Sol. $\sqrt{1 + \frac{x}{144}} = \frac{13}{12} \Rightarrow \left(1 + \frac{x}{144}\right) = \left(\frac{13}{12}\right)^2 = \frac{169}{144} \Rightarrow \frac{x}{144} = \frac{169}{144} - 1$
 $\Rightarrow \frac{x}{144} = \frac{25}{144} \Rightarrow x = 25.$

Ex. 13. Simplify : $\frac{1}{\sqrt{100} - \sqrt{99}} - \frac{1}{\sqrt{99} - \sqrt{98}} + \frac{1}{\sqrt{98} - \sqrt{97}} - \frac{1}{\sqrt{97} - \sqrt{96}} + \dots + \frac{1}{\sqrt{2} - \sqrt{1}}.$ (Section Officers', 2005)

Sol. Given expression

$$\begin{aligned} &= \frac{1}{\sqrt{100} - \sqrt{99}} \times \frac{\sqrt{100} + \sqrt{99}}{\sqrt{100} + \sqrt{99}} - \frac{1}{\sqrt{99} - \sqrt{98}} \times \frac{\sqrt{99} + \sqrt{98}}{\sqrt{99} + \sqrt{98}} + \frac{1}{\sqrt{98} - \sqrt{97}} \times \frac{\sqrt{98} + \sqrt{97}}{\sqrt{98} + \sqrt{97}} \\ &\quad - \frac{1}{\sqrt{97} - \sqrt{96}} \times \frac{\sqrt{97} + \sqrt{96}}{\sqrt{97} + \sqrt{96}} + \dots + \frac{1}{\sqrt{2} - \sqrt{1}} \times \frac{\sqrt{2} + \sqrt{1}}{\sqrt{2} + \sqrt{1}} \\ &= \frac{\sqrt{100} + \sqrt{99}}{(100 - 99)} - \frac{\sqrt{99} + \sqrt{98}}{(99 - 98)} + \frac{\sqrt{98} + \sqrt{97}}{(98 - 97)} - \frac{\sqrt{97} + \sqrt{96}}{(97 - 96)} + \dots + \frac{\sqrt{2} + \sqrt{1}}{(2 - 1)} \\ &= (\sqrt{100} + \sqrt{99}) - (\sqrt{99} + \sqrt{98}) + (\sqrt{98} + \sqrt{97}) - (\sqrt{97} + \sqrt{96}) + \dots + (\sqrt{2} + \sqrt{1}) \\ &= \sqrt{100} + \sqrt{1} = 10 + 1 = 11. \end{aligned}$$

Ex. 14. Find the sum : $3 + \frac{1}{\sqrt{3}} + \frac{1}{3 + \sqrt{3}} - \frac{1}{3 - \sqrt{3}}.$ (S.S.C., 2007)

Sol. $3 + \frac{1}{3} + \frac{1}{3 + \sqrt{3}} - \frac{1}{3 - \sqrt{3}} = 3 + \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} + \frac{1}{3 + \sqrt{3}} \times \frac{3 - \sqrt{3}}{3 - \sqrt{3}} - \frac{1}{3 - \sqrt{3}} \times \frac{3 + \sqrt{3}}{3 + \sqrt{3}}$
 $= 3 + \frac{\sqrt{3}}{3} + \frac{3 - \sqrt{3}}{(9 - 3)} - \frac{3 + \sqrt{3}}{(9 - 3)} = 3 + \frac{\sqrt{3}}{3} + \frac{3 - \sqrt{3}}{6} - \frac{3 + \sqrt{3}}{6}$
 $= \frac{18 + 2\sqrt{3} + 3 - \sqrt{3} - 3 - \sqrt{3}}{6} = \frac{18}{6} = 3.$

Ex. 15. Find the value of $\sqrt{3}$ upto three places of decimal.

Sol.
$$\begin{array}{r} 1 \overline{) 3.000000} \quad (1.732 \\ \underline{1} \\ 27 \underline{200} \\ 189 \\ \underline{1100} \\ 1029 \\ \underline{7100} \\ 6924 \end{array} \quad \therefore \sqrt{3} = 1.732.$$

Ex. 16. If $\sqrt{3} = 1.732$, find the value of $\sqrt{192} - \frac{1}{2}\sqrt{48} - \sqrt{75}$ correct to 3 places of decimal. (S.S.C., 2004)

Sol. $\sqrt{192} - \frac{1}{2}\sqrt{48} - \sqrt{75} = \sqrt{64 \times 3} - \frac{1}{2}\sqrt{16 \times 3} - \sqrt{25 \times 3} = 8\sqrt{3} - \frac{1}{2} \times 4\sqrt{3} - 5\sqrt{3}$
 $= 3\sqrt{3} - 2\sqrt{3} = \sqrt{3} = 1.732.$

Ex. 17. If $\sqrt{0.05 \times 0.5 \times a} = 0.5 \times 0.05 \times \sqrt{b}$, then find the value of $\frac{a}{b}$. (M.B.A., 2006)

Sol. Clearly, we have: $\frac{\sqrt{a}}{\sqrt{b}} = \frac{0.5 \times 0.05}{\sqrt{0.05 \times 0.5}} \Rightarrow \frac{a}{b} = \frac{0.5 \times 0.05 \times 0.5 \times 0.05}{0.05 \times 0.5} = 0.5 \times 0.05 = 0.025.$

Ex. 18. Evaluate : $\sqrt{\frac{9.5 \times .0085 \times 18.9}{.0017 \times 1.9 \times 0.021}}$.

Sol. Given exp. = $\sqrt{\frac{9.5 \times .0085 \times 18.900}{.0017 \times 1.9 \times 0.021}}$.

Now, since the sum of decimal places in the numerator and denominator under the radical sign is the same, we remove the decimal.

$$\therefore \text{Given exp.} = \sqrt{\frac{95 \times 85 \times 18900}{17 \times 19 \times 21}} = \sqrt{5 \times 5 \times 900} = 5 \times 30 = 150.$$

Ex. 19. Simplify : $\sqrt{[(12.1)^2 - (8.1)^2] \div [(0.25)^2 + (0.25)(19.95)]}$.

(C.B.I., 2003)

Sol. Given exp. = $\sqrt{\frac{(12.1+8.1)(12.1-8.1)}{(0.25)(0.25+19.95)}} = \sqrt{\frac{20.2 \times 4}{0.25 \times 20.2}} = \sqrt{\frac{4}{0.25}} = \sqrt{\frac{400}{25}} = \sqrt{16} = 4.$

Ex. 20. If $x = 1 + \sqrt{2}$ and $y = 1 - \sqrt{2}$, find the value of $(x^2 + y^2)$.

Sol. $x^2 + y^2 = (1 + \sqrt{2})^2 + (1 - \sqrt{2})^2 = 2[(1)^2 + (\sqrt{2})^2] = 2 \times 3 = 6.$

Ex. 21. Evaluate $\sqrt{0.9}$ upto 3 places of decimal.

Sol.

$$\begin{array}{r} 9 \quad | \quad 0.900000 \quad (0.948) \\ \quad \quad 81 \\ \hline 184 \quad | \quad 900 \\ \quad \quad 736 \\ \hline 1888 \quad | \quad 16400 \\ \quad \quad 15104 \\ \hline \end{array} \quad \therefore \sqrt{0.9} = 0.948.$$

Ex. 22. Find the square root of $0.\dot{1}$.

Sol. $\sqrt{0.\dot{1}} = \sqrt{\frac{1}{9}} = \frac{1}{3} = 0.333 \dots = 0.\dot{3}.$

Ex. 23. If $\sqrt{15} = 3.88$, find the value of $\sqrt{\frac{5}{3}}$.

Sol. $\sqrt{\frac{5}{3}} = \sqrt{\frac{5 \times 3}{3 \times 3}} = \frac{\sqrt{15}}{3} = \frac{3.88}{3} = 1.2933 \dots = 1.29\bar{3}.$

Ex. 24. Find the least square number which is exactly divisible by 10, 12, 15 and 18.

Sol. L.C.M. of 10, 12, 15, 18 = 180. Now, $180 = 2 \times 2 \times 3 \times 3 \times 5 = 2^2 \times 3^2 \times 5$.

To make it a perfect square, it must be multiplied by 5.

$\therefore$ Required number = $(2^2 \times 3^2 \times 5^2) = 900.$

Ex. 25. Find the greatest number of five digits which is a perfect square.

Sol. Greatest number of 5 digits is 99999.

$$\begin{array}{r} 3 \quad | \quad 99999 \quad (316) \\ \quad \quad 9 \\ \hline 61 \quad | \quad 99 \\ \quad \quad 61 \\ \hline 626 \quad | \quad 3899 \\ \quad \quad 3756 \\ \hline \end{array} \quad \begin{array}{r} 143 \\ \hline \end{array}$$

$\therefore$ Required number = $(99999 - 143) = 99856.$

Ex. 26. Find the smallest number that must be added to 1780 to make it a perfect square.

Sol.

$$\begin{array}{r} 4 \overline{) 1780} \quad (42 \\ \underline{16} \\ 82 \\ \underline{164} \\ 16 \end{array}$$

$\therefore$ Number to be added = $(43)^2 - 1780 = 1849 - 1780 = 69$.

Ex. 27. If $\sqrt{2} = 1.4142$, find the value of $\frac{\sqrt{2}}{(2+\sqrt{2})}$.

Sol.
$$\frac{\sqrt{2}}{(2+\sqrt{2})} = \frac{\sqrt{2}}{(2+\sqrt{2})} \times \frac{(2-\sqrt{2})}{(2-\sqrt{2})} = \frac{2\sqrt{2}-2}{(4-2)} = \frac{2(\sqrt{2}-1)}{2} = (\sqrt{2}-1) = (1.4142-1) = 0.4142.$$

Ex. 28. If $x = \left(\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} \right)$ and $y = \left(\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}} \right)$, find the value of $(x^2 + y^2)$.

Sol.
$$x = \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}-\sqrt{3})} \times \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})} = \frac{(\sqrt{5}+\sqrt{3})^2}{(5-3)} = \frac{5+3+2\sqrt{15}}{2} = 4+\sqrt{15}.$$

$$y = \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})} = \frac{(\sqrt{5}-\sqrt{3})^2}{(5-3)} = \frac{5+3-2\sqrt{15}}{2} = 4-\sqrt{15}.$$

$\therefore x^2 + y^2 = (4+\sqrt{15})^2 + (4-\sqrt{15})^2 = 2[(4)^2 + (\sqrt{15})^2] = 2 \times 31 = 62.$

Ex. 29. Find the value of: $\sqrt{6+\sqrt{6+\sqrt{6+\dots}}}$

(S.S.C., 2006)

Sol. Let $\sqrt{6+\sqrt{6+\sqrt{6+\dots}}} = x.$

Then, $\sqrt{6+x} = x \Leftrightarrow 6+x = x^2 \Rightarrow x^2 - x - 6 = 0 \Leftrightarrow x^2 - 3x + 2x - 6 = 0$

$\Leftrightarrow x(x-3) + 2(x-3) = 0 \Leftrightarrow (x-3)(x+2) = 0 \Leftrightarrow x = 3.$

$[\because x \neq -2]$

Hence, $\sqrt{6+\sqrt{6+\sqrt{6+\dots}}} = 3.$

Ex. 30. Find the cube root of 2744.

Sol. Method : Resolve the given number as the product of prime factors and take the product of prime factors, choosing one out of three of the same prime factors. Resolving 2744 as the product of prime factors, we get :

$$2744 = 2^3 \times 7^3.$$

$$\therefore \sqrt[3]{2744} = 2 \times 7 = 14.$$

2	2744
2	1372
2	686
7	343
7	49
	7

Ex. 31. By what least number 4320 be multiplied to obtain a number which is a perfect cube?

Sol. Clearly, $4320 = 2^3 \times 3^3 \times 2^2 \times 5.$

To make it a perfect cube, it must be multiplied by 2×5^2 i.e., 50.

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (✓) against the correct answer:

1. $\sqrt{53824} = ?$
 (a) 202 (b) 232
 (c) 242 (d) 332
2. The square root of 41209 is equal to (L.I.C.A.D.O., 2008)
 (a) 103 (b) 203
 (c) 303 (d) 403
3. The square root of 123454321 is (P.C.S., 2008)
 (a) 111111 (b) 12341
 (c) 11111 (d) 11211
4. The number of digits in the square root of 625685746009 is (S.S.C., 2007)
 (a) 4 (b) 5
 (c) 6 (d) 7
5. $\sqrt{\sqrt{17956} + \sqrt{24025}} = ?$ (Bank P.O., 2008)
 (a) 19 (b) 155
 (c) 256 (d) 289
 (e) None of these
6. $\sqrt{\sqrt{44944} + \sqrt{52441}} = ?$ (Bank P.O., 2008)
 (a) 17 (b) 312
 (c) 441 (d) 485
 (e) None of these
7. One-fourth of the sum of prime numbers, greater than 4 but less than 16, is the square of (P.C.S., 2009)
 (a) 3 (b) 4
 (c) 5 (d) 7
8. The value of $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}$ is
 (a) 4 (b) 6
 (c) 8 (d) 10
9. Evaluate $\sqrt{41 - \sqrt{21 + \sqrt{19 - \sqrt{9}}}}$.
 (a) 3 (b) 5
 (c) 6 (d) 6.4
10. $\sqrt{176 + \sqrt{2401}}$ is equal to
 (a) 14 (b) 15
 (c) 18 (d) 24
11. $\frac{\sqrt{196}}{7} \times \frac{\sqrt{441}}{7} \times \frac{120}{\sqrt{225}} = ?$ (R.R.B., 2008)
 (a) 48 (b) 54
 (c) 58 (d) 84
12. $\left(\sqrt{\frac{225}{729}} - \sqrt{\frac{25}{144}} \right) + \sqrt{\frac{16}{81}} = ?$
 (a) $\frac{1}{48}$ (b) $\frac{5}{48}$
 (c) $\frac{5}{16}$ (d) None of these
13. $(15)^2 + (18)^2 - 20 = \sqrt{?}$ (Bank P.O., 2006)
 (a) 22 (b) 23
 (c) 529 (d) 279841
 (e) None of these
14. $\sqrt{?} \times \sqrt{484} = 1034$ (L.I.C.A.D.O., 2007)
 (a) 2025 (b) 2209
 (c) 2304 (d) 2401
 (e) None of these
15. $\sqrt{1881} \times \sqrt{?} = 10137$ (Bank Recruitment, 2008)
 (a) 8281 (b) 8649
 (c) 9216 (d) 9409
 (e) None of these
16. In the equation $\frac{4050}{\sqrt{x}} = 450$, the value of x is (L.I.C.A.D.O., 2008)
 (a) 9 (b) 49
 (c) 81 (d) 100
17. $\sqrt{\frac{16}{25}} \times \sqrt{\frac{?}{25}} \times \frac{16}{25} = \frac{256}{625}$ (R.R.B. 2008)
 (a) 5 (b) 8
 (c) 16 (d) None of these
18. The square root of $(272^2 - 128^2)$ is
 (a) 144 (b) 200
 (c) 240 (d) 256
19. If $x * y = x + y + \sqrt{xy}$, the value of $6 * 24$ is
 (a) 41 (b) 42
 (c) 43 (d) 44
20. If $y = 5$, then what is the value of $10y\sqrt{y^3 - y^2}$?
 (a) $50\sqrt{2}$ (b) 100
 (c) $200\sqrt{5}$ (d) 500
21. $\sqrt{110\frac{1}{4}} = ?$
 (a) 10.25 (b) 10.5
 (c) 11.5 (d) 19.5

22. $\sqrt{\frac{25}{81} - \frac{1}{9}} = ?$ (Hotel Management, 2002)
- (a) $\frac{2}{3}$ (b) $\frac{4}{9}$
(c) $\frac{16}{81}$ (d) $\frac{25}{81}$
23. $[(\sqrt{81})^2]^2 = (?)^2$ (Specialist Officers, 2006)
- (a) 8 (b) 9
(c) 4096 (d) 6561
(e) None of these
24. The digit in the unit's place in the square root of 15876 is
- (a) 2 (b) 4
(c) 6 (d) 8
25. Which of the following is closest to $\sqrt{3}$? (S.S.C., 2005)
- (a) 1.69 (b) $\frac{173}{100}$
(c) 1.75 (d) $\frac{9}{5}$
26. How many two-digit numbers satisfy this property: The last digit (unit's digit) of the square of the two-digit number is 8?
- (a) 1 (b) 2
(c) 3 (d) None of these
27. What percentage of the numbers from 1 to 50 have squares that end in the digit 1? (M.B.A., 2006)
- (a) 1 (b) 5
(c) 10 (d) 11
(e) 20
28. While solving a mathematical problem, Samidha squared a number and then subtracted 25 from it rather than the required i.e., first subtracting 25 from the number and then squaring it. But she got the right answer. What was the given number? (Bank P.O., 2006)
- (a) 13 (b) 38
(c) 48
(d) Cannot be determined (e) None of these
29. How many perfect squares lie between 120 and 300? (S.S.C., 2010)
- (a) 5 (b) 6
(c) 7 (d) 8
30. The number of perfect square numbers between 50 and 1000 is
- (a) 21 (b) 22
(c) 23 (d) 24 (Section Officers', 2003)
31. A man born in the first half of the nineteenth century was x years old in the year x^2 . He was born in (M.B.A., 2011)
- (a) 1806 (b) 1812
(c) 1825 (d) 1836
32. R is a positive number. It is multiplied by 8 and then squared. The square is now divided by 4 and the square root is taken. The result of the square root is Q . What is the value of Q ? (SNAP, 2010)
- (a) $3R$ (b) $4R$
(c) $7R$ (d) $9R$
33. The smallest natural number which is a perfect square and which ends in 3 identical digits lies between
- (a) 1000 and 2000 (b) 2000 and 3000
(c) 3000 and 4000 (d) 4000 and 5000
34. $\left(\sqrt{2} + \frac{1}{\sqrt{2}}\right)^2$ is equal to (S.S.C., 2005)
- (a) $2\frac{1}{2}$ (b) $3\frac{1}{2}$
(c) $4\frac{1}{2}$ (d) $5\frac{1}{2}$
35. If the product of four consecutive natural numbers increased by a natural number p , is a perfect square, then the value of p is (C.P.O., 2006)
- (a) 1 (b) 2
(c) 4 (d) 8
36. What is the square root of 0.16?
- (a) 0.004 (b) 0.04
(c) 0.4 (d) 4
37. The value of $\sqrt{0.000441}$ is (S.S.C., 2002)
- (a) 0.00021 (b) 0.0021
(c) 0.021 (d) 0.21
38. $\sqrt{0.00004761}$ equals (C.B.I., 2003)
- (a) 0.00069 (b) 0.0069
(c) 0.0609 (d) 0.069
39. $1.5^2 \times \sqrt{0.0225} = ?$ (Bank P.O., 2002)
- (a) 0.0375 (b) 0.3375
(c) 3.275 (d) 32.75
40. $\sqrt{0.01} + \sqrt{0.0064} = ?$
- (a) 0.03 (b) 0.3
(c) 0.42 (d) None of these
41. The value of $\sqrt{0.01} + \sqrt{0.81} + \sqrt{1.21} + \sqrt{0.0009}$ is (S.S.C., 2002)
- (a) 2.03 (b) 2.1
(c) 2.11 (d) 2.13

42. $\sqrt{.0025} \times \sqrt{2.25} \times \sqrt{.0001} = ?$
 (a) .000075 (b) .0075
 (c) .075 (d) None of these
43. $\sqrt{1.5625} = ?$ (S.B.I.P.O., 2003)
 (a) 1.05 (b) 1.25
 (c) 1.45 (d) 1.55
44. If $\sqrt{.00000676} = .0026$, the square root of 67,60,000 is:
 (a) $\frac{1}{26}$ (b) 26
 (c) 260 (d) 2600
45. If $\sqrt{18225} = 135$, then the value of $(\sqrt{182.25} + \sqrt{1.8225} + \sqrt{0.018225} + \sqrt{0.00018225})$ is
 (a) 1.49985 (b) 14.9985
 (c) 149.985 (d) 1499.85
46. If $\sqrt{4096} = 64$, then the value of $\sqrt{40.96} + \sqrt{0.4096} + \sqrt{0.004096} + \sqrt{0.00004096}$ up to two places of decimals is (S.S.C., 2005)
 (a) 7.09 (b) 7.10
 (c) 7.11 (d) 7.12
47. Given that $\sqrt{13} = 3.605$ and $\sqrt{130} = 11.40$, find the value of $\sqrt{1.3} + \sqrt{1300} + \sqrt{0.013}$.
 (a) 36.164 (b) 36.304
 (c) 37.164 (d) 37.304
48. If $\frac{52}{x} = \sqrt{\frac{169}{289}}$, the value of x is
 (a) 52 (b) 58
 (c) 62 (d) 68
49. For what value of * the statement $\left(\frac{*}{15}\right)\left(\frac{*}{135}\right) = 1$ is true? (S.S.C., 2002)
 (a) 15 (b) 25
 (c) 35 (d) 45
50. Which number should replace both the question marks in the following equation? (Bank P.O., 2008)
 $\frac{?}{1776} = \frac{111}{?}$
 (a) 343 (b) 414
 (c) 644 (d) 543
 (e) None of these
51. Which number can replace both the question marks in the equation $\frac{4\frac{1}{2}}{?} = \frac{?}{32}$.
 (a) 1 (b) 7
 (c) $7\frac{1}{2}$ (d) None of these
52. What should come in place of both the question marks in the equation $\frac{?}{\sqrt{128}} = \frac{\sqrt{162}}{?}$.
 (a) 12 (b) 14
 (c) 144 (d) 196
53. If $\sqrt{x + \frac{x}{y}} = x\sqrt{\frac{x}{y}}$, where x and y are positive real numbers, then y is equal to (Hotel Mgmt, 2010)
 (a) $x + 1$ (b) $x - 1$
 (c) $x^2 + 1$ (d) $x^2 - 1$
54. The number $25^{64} \times 64^{25}$ is the square of a natural number n . The sum of the digits of n is (A.A.O. Exam., 2010)
 (a) 7 (b) 14
 (c) 21 (d) 28
55. If $0.13 \div p^2 = 13$, then p equals
 (a) 0.01 (b) 0.1
 (c) 10 (d) 100
56. What number should be divided by $\sqrt{0.25}$ to give the result as 25? (C.B.I., 2003)
 (a) 12.5 (b) 25
 (c) 50 (d) 125
57. If $\sqrt{3^n} = 729$, then the value of n is (Section Officers, 2003)
 (a) 6 (b) 8
 (c) 10 (d) 12
58. If $\sqrt{18 \times 14 \times x} = 84$, then x equals
 (a) 22 (b) 24
 (c) 28 (d) 32
59. $28\sqrt{?} + 1426 = \frac{3}{4}$ of 2872
 (a) 576 (b) 676
 (c) 1296 (d) 1444
60. $\sqrt{\frac{?}{169}} = \frac{54}{39}$
 (a) 108 (b) 324
 (c) 2916 (d) 4800
61. If $\sqrt{x} + \sqrt{441} = 0.02$, then the value of x is
 (a) 0.1764 (b) 1.764
 (c) 1.64 (d) 2.64
62. $\sqrt{\frac{.0196}{?}} = 0.2$
 (a) 0.49 (b) 0.7
 (c) 4.9 (d) None of these
63. $\sqrt{0.0169 \times ?} = 1.3$ (Hotel Management, 2001)
 (a) 10 (b) 100
 (c) 1000 (d) None of these

64. If $\sqrt{1369} + \sqrt{.0615 + x} = 37.25$, then x is equal to
 (a) 10^{-1} (b) 10^{-2}
 (c) 10^{-3} (d) None of these
65. If $\sqrt{(x-1)(y+2)} = 7$, x and y being positive whole numbers, then the values of x and y respectively are
 (a) 8, 5 (b) 15, 12
 (c) 22, 19 (d) None of these
66. If $\sqrt{.04 \times .4 \times a} = .004 \times .4 \times \sqrt{b}$, then $\frac{a}{b}$ is
 (a) 16×10^{-3} (b) 16×10^{-4}
 (c) 16×10^{-5} (d) None of these
67. Three-fifth of the square of a certain number is 126.15. What is the number?
 (a) 14.5 (b) 75.69
 (c) 145 (d) 210.25
68. $\sqrt[3]{\frac{0.361}{0.00169}} = ?$
 (a) $\frac{1.9}{13}$ (b) $\frac{19}{13}$
 (c) $\frac{1.9}{130}$ (d) $\frac{190}{13}$
69. $\sqrt[3]{\frac{48.4}{0.289}}$ is equal to (S.S.C., 2004)
 (a) $1\frac{5}{17}$ (b) $12\frac{1}{17}$
 (c) $12\frac{16}{17}$ (d) $129\frac{7}{17}$
70. If $\sqrt{1 + \frac{x}{169}} = \frac{14}{13}$, then x is equal to
 (a) 1 (b) 13
 (c) 27 (d) None of these
71. $\sqrt{1 + \frac{55}{729}} = 1 + \frac{x}{27}$, then the value of x is (C.D.S., 2003)
 (a) 1 (b) 3
 (c) 5 (d) 7
72. $\sqrt{\frac{4}{3}} - \sqrt{\frac{3}{4}} = ?$ (R.R.B., 2005)
 (a) $\frac{4\sqrt{3}}{6}$ (b) $\frac{1}{2\sqrt{3}}$
 (c) 1 (d) $-\frac{1}{2\sqrt{3}}$
73. The value of $\sqrt{2}$ upto three places of decimal is
 (a) 1.410 (b) 1.412
 (c) 1.413 (d) 1.414
74. $(2\sqrt{27} - \sqrt{75} + \sqrt{12})$ is equal to
 (a) $\sqrt{3}$ (b) $2\sqrt{3}$
 (c) $3\sqrt{3}$ (d) $4\sqrt{3}$
75. By how much does $\sqrt{12} + \sqrt{18}$ exceed $\sqrt{3} + \sqrt{2}$?
 (a) $\sqrt{2} - 4\sqrt{3}$ (b) $\sqrt{3} + 2\sqrt{2}$
 (c) $2(\sqrt{3} - \sqrt{2})$ (d) $3(\sqrt{3} - \sqrt{2})$
76. $\frac{\sqrt{24} + \sqrt{216}}{\sqrt{96}} = ?$
 (a) $2\sqrt{6}$ (b) 2
 (c) $6\sqrt{2}$ (d) $\frac{2}{\sqrt{6}}$
77. The value of $\frac{\sqrt{80} - \sqrt{112}}{\sqrt{45} - \sqrt{63}}$ is
 (a) $\frac{3}{4}$ (b) $1\frac{1}{3}$
 (c) $1\frac{7}{9}$ (d) $1\frac{3}{4}$
78. If $3\sqrt{5} + \sqrt{125} = 17.88$, then what will be the value of $\sqrt{80} + 6\sqrt{5}$?
 (a) 13.41 (b) 20.46
 (c) 21.66 (d) 22.35
79. $\sqrt{50} \times \sqrt{98}$ is equal to
 (a) 63.75 (b) 65.95
 (c) 70 (d) 70.25
80. Given $\sqrt{2} = 1.414$. The value of $\sqrt{8} + 2\sqrt{32} - 3\sqrt{128} + 4\sqrt{50}$ is: (S.S.C., 2003)
 (a) 8.426 (b) 8.484
 (c) 8.526 (d) 8.876
81. The approximate value of $\frac{3\sqrt{12}}{2\sqrt{28}} + \frac{2\sqrt{21}}{\sqrt{98}}$ is (Section Officers, 2003)
 (a) 1.0605 (b) 1.0727
 (c) 1.6007 (d) 1.6026
82. $\sqrt{110.25} \times \sqrt{0.01} + \sqrt{0.0025} - \sqrt{420.25}$ equals (SNAP, 2010)
 (a) 0.50 (b) 0.64
 (c) 0.73 (d) 0.75
83. $\sqrt{\frac{.081 \times .484}{.0064 \times 6.25}}$ is equal to
 (a) 0.9 (b) 0.99
 (c) 9 (d) 99

84. $\sqrt{\frac{0.204 \times 42}{0.07 \times 3.4}}$ is equal to
 (a) $\frac{1}{6}$ (b) 0.06
 (c) 0.6 (d) 6
85. $\sqrt{\frac{0.081 \times 0.324 \times 4.624}{1.5625 \times 0.0289 \times 72.9 \times 64}}$ is equal to
 (a) 0.024 (b) 0.24
 (c) 2.4 (d) 24
86. $\sqrt{\frac{9.5 \times .085}{.0017 \times .19}}$ equals
 (a) .05 (b) 5
 (c) 50 (d) 500
87. The value of $\sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{(0.003)^2 + (0.021)^2 + (0.0065)^2}}$ is
 (a) 0.1 (b) 10
 (c) 10^2 (d) 10^3 (S.S.C., 2002)
88. The square root of $(7 + 3\sqrt{5})(7 - 3\sqrt{5})$ is (S.S.C., 2004)
 (a) $\sqrt{5}$ (b) 2
 (c) 4 (d) $3\sqrt{5}$
89. $\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)^2$ simplifies to
 (a) $\frac{3}{4}$ (b) $\frac{4}{\sqrt{3}}$
 (c) $\frac{4}{3}$ (d) None of these
90. If $a = 0.1039$, then the value of $\sqrt{4a^2 - 4a + 1} + 3a$ is
 (a) 0.1039 (b) 0.2078
 (c) 1.1039 (d) 2.1039 (C.B.I., 2003)
91. The square root of $\frac{(0.75)^3}{1 - 0.75} + [0.75 + (0.75)^2 + 1]$ is
 (a) 1 (b) 2
 (c) 3 (d) 4
92. If $3a = 4b = 6c$ and $a + b + c = 27\sqrt{29}$, then $\sqrt{a^2 + b^2 + c^2}$ is
 (a) $3\sqrt{29}$ (b) 81
 (c) 87 (d) None of these
93. The square root of $0.\bar{4}$ is (S.S.C., 2004)
 (a) $0.\bar{6}$ (b) $0.\bar{7}$
 (c) $0.\bar{8}$ (d) $0.\bar{9}$
94. Which one of the following numbers has rational square root?
 (a) 0.4 (b) 0.09
 (c) 0.9 (d) 0.025
95. The value of $\sqrt{0.4}$ is
 (a) 0.02 (b) 0.2
 (c) 0.51 (d) 0.63
96. $\sqrt{0.2} = ?$ (R.R.B., 2007)
 (a) 0.02 (b) 0.2
 (c) 0.447 (d) 0.632
97. The value of $\sqrt{0.121}$ is
 (a) 0.011 (b) 0.11
 (c) 0.347 (d) 1.1
98. The value of $\sqrt{0.064}$ is
 (a) 0.008 (b) 0.08
 (c) 0.252 (d) 0.8
99. The value of $\sqrt{\frac{0.16}{0.4}}$ is (IGNOU, 2003)
 (a) 0.02 (b) 0.2
 (c) 0.63 (d) None of these
100. The value of $\frac{1 + \sqrt{0.01}}{1 - \sqrt{0.1}}$ is close to
 (a) 0.6 (b) 1.1
 (c) 1.6 (d) 1.7
101. The square root of 535.9225 is (R.R.B., 2006)
 (a) 23.15 (b) 23.45
 (c) 24.15 (d) 28.25
102. If $\sqrt{5} = 2.236$, then the value of $\frac{1}{\sqrt{5}}$ is
 (a) .367 (b) .447
 (c) .745 (d) None of these
103. If $\sqrt{24} = 4.899$, the value of $\sqrt{\frac{8}{3}}$ is
 (a) 0.544 (b) 1.333
 (c) 1.633 (d) 2.666
104. If $\sqrt{6} = 2.449$, then the value of $\frac{3\sqrt{2}}{2\sqrt{3}}$ is
 (a) 0.6122 (b) 0.8163
 (c) 1.223 (d) 1.2245
105. If $\sqrt{5} = 2.236$, then the value of $\frac{\sqrt{5}}{2} - \frac{10}{\sqrt{5}} + \sqrt{125}$ is equal to
 (a) 5.59 (b) 7.826
 (c) 8.944 (d) 10.062

106. If $2 * 3 = \sqrt{13}$ and $3 * 4 = 5$, then the value of $5 * 12$ is
 (a) $\sqrt{17}$ (b) $\sqrt{29}$
 (c) 12 (d) 13
107. If $1537*$ is a perfect square, then the digit which replaces * is (Hotel Management, 2007)
 (a) 2 (b) 4
 (c) 5 (d) 6
108. The smallest perfect square that is divisible by $7!$ is (I.I.F.T., 2010)
 (a) 19600 (b) 44100
 (c) 176400 (d) 705600
109. The least perfect square number divisible by 3, 4, 5, 6 and 8 is
 (a) 900 (b) 1200
 (c) 2500 (d) 3600
110. The least perfect square, which is divisible by each of 21, 36 and 66, is
 (a) 213444 (b) 214344
 (c) 214434 (d) 231444
111. The least number by which 294 must be multiplied to make it a perfect square, is
 (a) 2 (b) 3
 (c) 6 (d) 24
112. Find the smallest number by which 5808 should be multiplied so that the product becomes a perfect square.
 (a) 2 (b) 3
 (c) 7 (d) 11
113. The least number by which 1470 must be divided to get a number which is a perfect square, is
 (a) 5 (b) 6
 (c) 15 (d) 30
114. What is the smallest number to be subtracted from 549162 in order to make it a perfect square?
 (a) 28 (b) 36
 (c) 62 (d) 81
115. What is the least number which should be subtracted from 0.000326 to make it a perfect square?
 (a) 0.000002 (b) 0.000004
 (c) 0.02 (d) 0.04
116. What is the least number to be added to 7700 to make it a perfect square?
 (a) 77 (b) 98
 (c) 131 (d) 221
 (e) None of these (Bank Recruitment, 2008)
117. The smallest number to be added to 680621 to make the sum a perfect square is (S.S.C., 2005)
 (a) 4 (b) 5
 (c) 6 (d) 8
118. The greatest four-digit perfect square number is
 (a) 9000 (b) 9801
 (c) 9900 (d) 9981
119. The least number of 4 digits which is a perfect square is
 (a) 1000 (b) 1016
 (c) 1024 (d) 1036
120. The sum of 18 consecutive natural numbers is a perfect square. What is the smallest possible value of this sum? (A.A.O. Exam, 2009)
 (a) 169 (b) 225
 (c) 289 (d) 441
121. $\sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{3}}}}$
 $\cdot \sqrt{2-\sqrt{2+\sqrt{2+\sqrt{3}}}}$ is equal to
 (a) 1 (b) 2
 (c) 4 (d) $\sqrt{6}$
122. Given $\sqrt{5} = 2.2361$, $\sqrt{3} = 1.7321$, then $\frac{1}{\sqrt{5}-\sqrt{3}}$ is equal to
 (a) 1.98 (b) 1.984
 (c) 1.9841 (d) 2
123. $\frac{1}{(\sqrt{9}-\sqrt{8})} - \frac{1}{(\sqrt{8}-\sqrt{7})} + \frac{1}{(\sqrt{7}-\sqrt{6})} - \frac{1}{(\sqrt{6}-\sqrt{5})}$
 $+ \frac{1}{(\sqrt{5}-\sqrt{4})}$ is equal to (M.B.A., 2007)
 (a) 0 (b) $\frac{1}{3}$
 (c) 1 (d) 5
124. Determine the value of
 $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \dots + \frac{1}{\sqrt{120}+\sqrt{121}}$.
 (a) 8 (b) 10
 (c) $\sqrt{120}$ (d) $12\sqrt{2}$ (M.A.T., 2005)
125. The expression $1 - \frac{1}{1+\sqrt{3}} + \frac{1}{1-\sqrt{3}}$ equals
 (a) $1-\sqrt{3}$ (b) 1
 (c) $-\sqrt{3}$ (d) $\sqrt{3}$ (M.B.A., 2011)
126. $\left(2 + \sqrt{2} + \frac{1}{2+\sqrt{2}} + \frac{1}{\sqrt{2}-2}\right)$ simplifies to
 (a) $2-\sqrt{2}$ (b) 2
 (c) $2+\sqrt{2}$ (d) $2\sqrt{2}$ (M.B.A., 2007)

127. What is the value of $\frac{1}{\sqrt{5}+\sqrt{3}} + \frac{2}{3+\sqrt{5}} - \frac{3}{3+\sqrt{3}}$? (I.I.F.T., 2005)
- (a) $-\frac{1}{2}$ (b) 0
(c) $\frac{1}{2}$ (d) 1
128. If $\sqrt{2} = 1.4142$, the value of $\frac{7}{(3+\sqrt{2})}$ is (R.R.B., 2005)
- (a) 1.5858 (b) 3.4852
(c) 3.5858 (d) 4.4142
129. $\left[\frac{3\sqrt{2}}{\sqrt{6}-\sqrt{3}} - \frac{4\sqrt{3}}{\sqrt{6}-\sqrt{2}} - \frac{6}{\sqrt{8}-\sqrt{12}} \right] = ?$ (Teachers' Exam, 2010)
- (a) $\sqrt{3}-\sqrt{2}$ (b) $\sqrt{3}+\sqrt{2}$
(c) $5\sqrt{3}$ (d) 1
130. $\frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}}$ is equal to
- (a) 1 (b) 2
(c) $6-\sqrt{35}$ (d) $6+\sqrt{35}$
131. If $\frac{5+2\sqrt{3}}{7+4\sqrt{3}} = a+b\sqrt{3}$, then (R.R.B., 2010)
- (a) $a = -11, b = -6$ (b) $a = -11, b = 6$
(c) $a = 11, b = -6$ (d) $a = 6, b = 11$
132. If $\sqrt{2} = 1.414$, the square root of $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ is nearest to
- (a) 0.172 (b) 0.414
(c) 0.586 (d) 1.414
133. Given that $\sqrt{3} = 1.732$, the value of $\frac{3+\sqrt{6}}{5\sqrt{3}-2\sqrt{12}-\sqrt{32}+\sqrt{50}}$ is (S.S.C., 2007)
- (a) 1.414 (b) 1.732
(c) 2.551 (d) 4.899
134. $\left(\frac{2+\sqrt{3}}{2-\sqrt{3}} + \frac{2-\sqrt{3}}{2+\sqrt{3}} + \frac{\sqrt{3}-1}{\sqrt{3}+1} \right)$ simplifies to (S.S.C., 2008)
- (a) $16-\sqrt{3}$ (b) $4-\sqrt{3}$
(c) $2-\sqrt{3}$ (d) $2+\sqrt{3}$
135. If $x = \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$ and $y = \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$, then $(x+y)$ equals (S.S.C., 2005)
- (a) $2(\sqrt{5}+\sqrt{3})$ (b) $2\sqrt{15}$
(c) 8 (d) 16
136. $\frac{\sqrt{2}(2+\sqrt{3})}{\sqrt{3}(\sqrt{3}+1)} \times \frac{\sqrt{2}(2-\sqrt{3})}{\sqrt{3}(\sqrt{3}-1)}$ is equal to (C.P.O., 2006)
- (a) $3\sqrt{2}$ (b) $\frac{\sqrt{2}}{3}$
(c) $\frac{2}{3}$ (d) $\frac{1}{3}$
137. $\frac{12}{3+\sqrt{5}+2\sqrt{2}}$ is equal to (S.S.C., 2007)
- (a) $1-\sqrt{5}+\sqrt{2}+\sqrt{10}$
(b) $1+\sqrt{5}+\sqrt{2}-\sqrt{10}$
(c) $1+\sqrt{5}-\sqrt{2}+\sqrt{10}$
(d) $1-\sqrt{5}-\sqrt{2}+\sqrt{10}$
138. $\left[\frac{1}{\sqrt{2}+\sqrt{3}-\sqrt{5}} + \frac{1}{\sqrt{2}-\sqrt{3}-\sqrt{5}} \right]$ in simplified form equals (S.S.C., 2005)
- (a) 0 (b) $\frac{1}{\sqrt{2}}$
(c) 1 (d) $\sqrt{2}$
139. If $x = (7-4\sqrt{3})$, then the value of $\left(x + \frac{1}{x} \right)$ is
- (a) $3\sqrt{3}$ (b) $8\sqrt{3}$
(c) 14 (d) $14+8\sqrt{3}$
140. If $x = 3+\sqrt{8}$, then $x^2 + \frac{1}{x^2}$ is equal to (S.S.C., 2007)
- (a) 30 (b) 34
(c) 36 (d) 38
141. If $a = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$, $b = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$, then the value of $a^2 + b^2$ would be (M.B.A., 2008)
- (a) 10 (b) 98
(c) 99 (d) 100
142. If $a = \frac{\sqrt{5}+1}{\sqrt{5}-1}$ and $b = \frac{\sqrt{5}-1}{\sqrt{5}+1}$, the value of $\left(\frac{a^2+ab+b^2}{a^2-ab+b^2} \right)$ is
- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$
(c) $\frac{3}{5}$ (d) $\frac{5}{3}$

143. If $x = \sqrt{1 + \sqrt{1 + \sqrt{1 + \dots \infty}}}$ then the positive value of x is (M.B.A., 2006)
- (a) $\frac{\sqrt{7} + 1}{2}$ (b) $\frac{\sqrt{6} + 1}{2}$
 (c) $\frac{\sqrt{5} + 1}{2}$ (d) $\frac{\sqrt{3} + 1}{2}$
144. $\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$ is equal to (S.S.C., 2005)
- (a) 1 (b) 1.5
 (c) 2 (d) 2.5
145. If $a = \sqrt{3 + \sqrt{3 + \sqrt{3 + \dots}}}$, then which of the following is true? (M.B.A., 2007)
- (a) $2 < a < 3$ (b) $a > 3$
 (c) $3 < a < 4$ (d) $a = 3$
146. $\frac{\sqrt{3+x} + \sqrt{3-x}}{\sqrt{3+x} - \sqrt{3-x}} = 2$. Then x is equal to (S.S.C., 2010)
- (a) $\frac{5}{7}$ (b) $\frac{7}{5}$
 (c) $\frac{5}{12}$ (d) $\frac{12}{5}$
147. One-fourth of a herd of camels was seen in the forest. Twice the square root of the herd had gone to mountains and the remaining 15 camels were seen on the bank of a river. Find the total number of camels. (M.A.T., 2005)
- (a) 32 (b) 34
 (c) 35 (d) 36
148. A gardener plants 17956 trees in such a way that there are as many rows as there are trees in a row. The number of trees in a row are (M.B.A., 2006)
- (a) 134 (b) 136
 (c) 144 (d) 154
149. The number of trees in each row of a garden is equal to the total number of rows in the garden. After 111 trees have been uprooted in a storm, there remain 10914 trees in the garden. The number of rows of trees in the garden is (C.P.O., 2007)
- (a) 100 (b) 105
 (c) 115 (d) 125
150. 1250 oranges were distributed among a group of girls of a class. Each girl got twice as many oranges as the number of girls in that group. The number of girls in the group was (P.C.S., 2006)
- (a) 25 (b) 45
 (c) 50 (d) 100
151. A General wishes to draw up his 36581 soldiers in the form of a solid square. After arranging them, he found that some of them are left over. How many are left?
- (a) 65 (b) 81
 (c) 100 (d) None of these
152. A group of students decided to collect as many paise from each member of the group as is the number of members. If the total collection amounts to ₹ 59.29, the number of members in the group is
- (a) 57 (b) 67
 (c) 77 (d) 87
153. A mobile company offered to pay the Indian Cricket Team as much money per run scored by the side as the total number it gets in a one-dayer against Australia. Which one of the following cannot be the total amount to be spent by the company in this deal? (P.C.S., 2008)
- (a) 21,904 (b) 56,169
 (c) 1,01,761 (d) 1,21,108
154. $\sqrt[3]{148877} = ?$ (L.I.C.A.D.O., 2007)
- (a) 43 (b) 49
 (c) 53 (d) 59
155. $\sqrt[3]{681472} = ?$ (Bank P.O., 2009)
- (a) 76 (b) 88
 (c) 96 (d) 98
156. $1728 \div \sqrt[3]{262144} \times ? - 288 = 4491$ (Bank P.O., 2008)
- (a) 148 (b) 156
 (c) 173 (d) 177
157. $99 \times 21 - \sqrt[3]{7} = 1968$ (NABARD, 2008)
- (a) 1367631 (b) 111
 (c) 1366731 (d) 1367
158. The cube root of .000216 is
- (a) .6 (b) .06
 (c) .006 (d) None of these
159. $\sqrt[3]{4 \frac{12}{125}} = ?$
- (a) $1 \frac{2}{5}$ (b) $1 \frac{3}{5}$
 (c) $1 \frac{4}{5}$ (d) $2 \frac{2}{5}$
160. $\sqrt[3]{\sqrt{.000064}} = ?$
- (a) .02 (b) .2
 (c) 2 (d) None of these
161. The smallest positive integer n , for which $864n$ is a perfect cube, is (C.P.O., 2007)
- (a) 1 (b) 2
 (c) 3 (d) 4
162. Value of $\sqrt{.01} \times \sqrt[3]{.008} - .02$ is (P.C.S., 2006)
- (a) 0 (b) 1
 (c) 2 (d) 3

- 163.** The value of $\sqrt[3]{\frac{0.2 \times 0.2 \times 0.2 + 0.04 \times 0.04 \times 0.04}{0.4 \times 0.4 \times 0.4 + 0.08 \times 0.08 \times 0.08}}$ is (S.S.C., 2005)
- (a) 0.125 (b) 0.25
(c) 0.5 (d) 0.75
- 164.** A rationalising factor of $(\sqrt[3]{9} - \sqrt[3]{3} + 1)$ is (S.S.C., 2007)
- (a) $\sqrt[3]{3} - 1$ (b) $\sqrt[3]{3} + 1$
(c) $\sqrt[3]{9} - 1$ (d) $\sqrt[3]{9} + 1$
- 165.** The largest four-digit number which is a perfect cube, is
- (a) 8000 (b) 9261
(c) 9999 (d) None of these
- 166.** By what least number must 21600 be multiplied so as to make it a perfect cube? (M.A.T., 2002)
- (a) 6 (b) 10
(c) 20 (d) 30
- 167.** What is the smallest number by which 3600 be divided to make it a perfect cube?
- (a) 9 (b) 50
(c) 300 (d) 450
- 168.** Which smallest number must be added to 710 so that the sum is a perfect cube? (S.S.C., 2005)
- (a) 11 (b) 19
(c) 21 (d) 29
- 169.** Solve $\sqrt{7921} = ?$ [Indian Railway Gr. 'D' Exam, 2014]
- (a) 89 (b) 87
(c) 37 (d) 47
- 170.** Solve $\sqrt[4]{(625)^3} = ?$ [Indian Railway Gr. 'D' Exam, 2014]
- (a) $\sqrt[3]{1875}$ (b) 25
(c) 125 (d) None of these
- 171.** If $\sqrt{y} = 4x$, then $\frac{x^2}{y}$ is [SSC—CHSL (10+2) Exam, 2015]
- (a) 2 (b) $\frac{1}{16}$
(c) $\frac{1}{4}$ (d) 4
- Direction (Q. No. 172):** What approximate value will come in place of question mark(?) in the given question? (You are not expected to calculate the exact value)
- 172.** $\sqrt{2025.11} \times \sqrt{256.04} + \sqrt{399.95} \times \sqrt{7} = 33.98 \times 40.11$ [IBPS—Bank Spl. Officer (IT) Exam, 2015]
- (a) 1682 (b) 1024
(c) 1582 (d) 678
- 173.** If $\sqrt{7} = 2.645$, then the value of $\frac{1}{\sqrt{28}}$ up to three places of decimal is [SSC—CHSL (10+2) Exam, 2015]
- (a) 0.183 (b) 0.185
(c) 0.187 (d) 0.189
- 174.** Solve: $\left(\sqrt{\frac{25}{9}} - \sqrt{\frac{64}{81}}\right) + \sqrt{\frac{16}{324}} = ?$
- (a) 4.5 (b) 2.5
(c) 1.5 (d) 3.5 [United India Insurance Co. Ltd. (UIICL) Assistant (Online) Exam, 2015]
- 175.** Solve $1728 \div \sqrt[3]{262144} \times ? - 288 = 4491$
- (a) 148 (b) 156
(c) 173 (d) 177
(e) 185 [United India Insurance Co. Ltd. (UIICL) Assistant (Online) Exam, 2015]
- 176.** Solve: $(\sqrt{7} + 11)^2 = (?)^{\frac{1}{3}} + 2\sqrt{847} + 122$
- (a) $36 + 44\sqrt{7}$ (b) 6
(c) 216 (d) 36 [IDBI Bank Executive Officers Exam, 2015]
- Directions (Q. No. 177 & 178):** What will come in place of question mark in these questions?
- 177.** $? - \sqrt{(784)} = 6 \times \sqrt{(324)}$ [NICL—AAO Exam, 2015]
- (a) 128 (b) 160
(c) 236 (d) 136
- 178.** $\sqrt{(2116)} - \sqrt{1600} = \sqrt{(?)}$ [NICL—AAO Exam, 2015]
- (a) 20 (b) 64
(c) 81 (d) 36
- 179.** Solve $\sqrt{(27 \div 5 \times ?)} + 15 = 5.4 \div 6 + 0.3$ [IBPS—RRB Office Assistant (Online) Exam, 2015]
- (a) 2 (b) 6
(c) 10 (d) 4
- 180.** $\sqrt{24 \div 0.5 + 1} + \sqrt{18 \div 0.6 + 6} = ?$
- (a) 19 (b) 13
(c) 12 (d) 15 [IBPS—RRB Office Assistant (Online) Exam, 2015]
- 181.** $(\sqrt{63} + \sqrt{252}) \times (\sqrt{175} + \sqrt{28}) = ?$ [IBPS—RRB Office Assistant (Online) Exam, 2015]
- (a) $16\sqrt{7}$ (b) 441
(c) 16 (d) $7\sqrt{7}$
- 182.** $9x^2 + 25 - 30x$ can be expressed as the square of
- (a) $3x^2 - 25$ (b) $3x - 5$
(c) $-3x - 5$ (d) $3x - 5$ [SSC—CHSL (10+2) Exam, 2015]
- 183.** If $\sqrt{33} = 5.745$, then the value of the following is approximately $\sqrt{\frac{3}{11}}$ [SSC—CHSL (10+2) Exam, 2015]
- (a) 1 (b) 6.32
(c) 0.5223 (d) 2.035

184. $\sqrt{?} + 14 = \sqrt{2601}$ [SBI Jr. Associates (Pre.) Exam, 2016]

- (a) 1521 (b) 1369
(c) 1225 (d) 961

185. If $a = \frac{\sqrt{3}}{2}$, then $\sqrt{1+a} + \sqrt{1-a} = ?$

[DMRC—Train Operator (Station Controller) Exam, 2016]

- (a) $(2 - \sqrt{3})$ (b) $(2 + \sqrt{3})$
(c) $\left(\frac{\sqrt{3}}{2}\right)$ (d) $\sqrt{3}$

186. What is $\frac{5 + \sqrt{10}}{5\sqrt{5} - 2\sqrt{20} - \sqrt{32} + \sqrt{50}}$ equal to? [CDS 2016]

- (a) 5 (b) $5\sqrt{2}$
(c) $5\sqrt{5}$ (d) $\sqrt{5}$

187. The square root of $\frac{(0.75)^3}{1-0.75} + [0.75 + (0.75)^2 + 1]$ is

- (a) 1 (b) 2 [CDS 2016]
(c) 3 (d) 4

188. $\sqrt{10 + 2\sqrt{6} + 2\sqrt{10} + 2\sqrt{15}}$ is equal to

- (a) $(\sqrt{2} + \sqrt{3} + \sqrt{5})$ (b) $(\sqrt{2} + \sqrt{3} - \sqrt{5})$
(c) $(\sqrt{2} + \sqrt{5} - \sqrt{3})$ (d) None of these

[DMRC—Customer Relationship Assistant (CRA) Exam, 2016]

ANSWERS

1. (b)	2. (b)	3. (c)	4. (c)	5. (e)	6. (e)	7. (a)	8. (a)	9. (c)	10. (b)
11. (a)	12. (c)	13. (d)	14. (b)	15. (b)	16. (c)	17. (c)	18. (c)	19. (b)	20. (d)
21. (b)	22. (b)	23. (e)	24. (c)	25. (b)	26. (d)	27. (e)	28. (a)	29. (c)	30. (d)
31. (a)	32. (b)	33. (a)	34. (c)	35. (a)	36. (c)	37. (c)	38. (b)	39. (b)	40. (b)
41. (d)	42. (d)	43. (b)	44. (d)	45. (b)	46. (c)	47. (d)	48. (d)	49. (d)	50. (e)
51. (d)	52. (a)	53. (d)	54. (b)	55. (b)	56. (a)	57. (d)	58. (c)	59. (b)	60. (b)
61. (a)	62. (a)	63. (b)	64. (c)	65. (a)	66. (c)	67. (a)	68. (d)	69. (c)	70. (c)
71. (a)	72. (b)	73. (d)	74. (c)	75. (b)	76. (b)	77. (b)	78. (d)	79. (c)	80. (b)
81. (a)	82. (a)	83. (b)	84. (d)	85. (a)	86. (c)	87. (b)	88. (b)	89. (c)	90. (c)
91. (b)	92. (c)	93. (a)	94. (b)	95. (d)	96. (c)	97. (c)	98. (c)	99. (c)	100. (c)
101. (a)	102. (b)	103. (c)	104. (d)	105. (b)	106. (d)	107. (d)	108. (c)	109. (d)	110. (a)
111. (c)	112. (b)	113. (d)	114. (d)	115. (a)	116. (e)	117. (a)	118. (b)	119. (c)	120. (b)
121. (a)	122. (c)	123. (d)	124. (b)	125. (a)	126. (b)	127. (b)	128. (a)	129. (c)	130. (d)
131. (c)	132. (b)	133. (b)	134. (a)	135. (c)	136. (d)	137. (b)	138. (b)	139. (c)	140. (b)
141. (b)	142. (b)	143. (c)	144. (c)	145. (a)	146. (d)	147. (d)	148. (a)	149. (b)	150. (a)
151. (c)	152. (c)	153. (d)	154. (c)	155. (b)	156. (d)	157. (a)	158. (b)	159. (b)	160. (b)
161. (b)	162. (a)	163. (c)	164. (b)	165. (b)	166. (b)	167. (d)	168. (b)	169. (a)	170. (c)
171. (b)	172. (b)	173. (d)	174. (d)	175. (d)	176. (c)	177. (d)	178. (d)	179. (d)	180. (b)
181. (b)	182. (d)	183. (c)	184. (b)	185. (d)	186. (d)	187. (b)	188. (a)		

SOLUTIONS

1.

$$\begin{array}{r} 2 \overline{) 5 \ 38 \ 24 \ (232)} \\ \underline{4} \\ 43 \overline{) 1 \ 38} \\ \underline{1 \ 29} \\ 462 \overline{) 9 \ 24} \\ \underline{9 \ 24} \\ 0 \\ \times \end{array}$$

$$\therefore \sqrt{53824} = 232.$$

2.

$$\begin{array}{r} 2 \overline{) 4 \ 12 \ 09 \ (203)} \\ \underline{4} \\ 40 \overline{) 12} \\ \underline{12} \\ 403 \overline{) 12 \ 09} \\ \underline{12 \ 09} \\ 0 \\ \times \end{array}$$

$$\therefore \sqrt{41209} = 203.$$

3.

$$\begin{array}{r} 1 \overline{) 1 \ 23 \ 45 \ 43 \ 21 \ (11111)} \\ \underline{1} \\ 21 \overline{) 23} \\ \underline{21} \\ 221 \overline{) 2 \ 45} \\ \underline{2 \ 21} \\ 2221 \overline{) 24 \ 43} \\ \underline{22 \ 21} \\ 22221 \overline{) 222 \ 21} \\ \underline{222 \ 21} \\ 0 \\ \times \end{array}$$

$$\therefore \sqrt{123454321} = 11111.$$

4. The number of digits in the square root of a perfect square number of n digits is

(i) $\frac{n}{2}$, if n is even

(ii) $\frac{n+1}{2}$, if n is odd

Here, $n = 12$. So, required number of digits = $\frac{n}{2} = \frac{12}{2} = 6$.

5. $\sqrt{17956} + \sqrt{24025} = \sqrt{134 + 155} = \sqrt{289} = 17$.

6. $\sqrt{44944} + \sqrt{52441} = \sqrt{212 + 229} = \sqrt{441} = 21$.

7. Sum of prime numbers greater than 4 but less than 16 = $(5 + 7 + 11 + 13) = 36$.

$\therefore \frac{1}{4} \times 36 = 9 = 3^2$.

8. Given exp. $= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + 15}}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{169}}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{108 + 13}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{121}}}$
 $= \sqrt{10 + \sqrt{25 + 11}} = \sqrt{10 + \sqrt{36}}$
 $= \sqrt{10 + 6} = \sqrt{16} = 4$.

9. Given exp. $= \sqrt{41 - \sqrt{21 + \sqrt{19 - 3}}}$
 $= \sqrt{41 - \sqrt{21 + \sqrt{16}}} = \sqrt{41 - \sqrt{21 + 4}}$
 $= \sqrt{41 - \sqrt{25}} = \sqrt{41 - 5} = \sqrt{36} = 6$.

10. Given exp. $= \sqrt{176 + 49} = \sqrt{225} = 15$.

11. Given exp. $= \frac{14}{7} \times \frac{21}{7} \times \frac{120}{15} = 2 \times 3 \times 8 = 48$.

12. Given exp. $= \left(\frac{\sqrt{225}}{\sqrt{729}} - \frac{\sqrt{25}}{\sqrt{144}} \right) + \frac{\sqrt{16}}{\sqrt{81}}$
 $= \left(\frac{15}{27} - \frac{5}{12} \right) + \frac{4}{9} = \left(\frac{15}{108} \times \frac{9}{4} \right) = \frac{5}{16}$.

13. Let $(15)^2 + (18)^2 - 20 = \sqrt{x}$.

Then, $\sqrt{x} = 225 + 324 - 20 = 529$

$\Leftrightarrow x = (529)^2 = 279841$.

14. Let $\sqrt{x} \times \sqrt{484} = 1034$. Then, $\sqrt{x} \times 22 = 1034$

$\Leftrightarrow \sqrt{x} = \frac{1034}{22} = 47$

$\Leftrightarrow x = (47)^2 = 2209$.

15. Let $\sqrt{11881} \times \sqrt{x} = 10137$. Then, $109 \times \sqrt{x} = 10137$

$\Leftrightarrow \sqrt{x} = \frac{10137}{109} = 93$

$\Leftrightarrow x = (93)^2 = 8649$.

16. $\frac{4050}{\sqrt{x}} = 450 \Leftrightarrow \sqrt{x} = \frac{4050}{450} = 9$

$\Leftrightarrow x = 9^2 = 81$.

17. Let $\sqrt{\frac{16}{25}} \times \sqrt{\frac{x}{25}} \times \frac{16}{25} = \frac{256}{625}$.

Then, $\frac{4}{5} \times \frac{\sqrt{x}}{2} \times \frac{16}{25} = \frac{256}{625}$

$\Leftrightarrow \frac{64\sqrt{x}}{625} = \frac{256}{625}$

$\Leftrightarrow \sqrt{x} = \frac{256}{625} \times \frac{625}{64}$

$\Leftrightarrow x = 4^2 = 16$.

18. $\sqrt{(272)^2 - (128)^2} = \sqrt{(272 + 128)(272 - 128)}$
 $= \sqrt{400 \times 144} = \sqrt{57600} = 240$.

19. $6 * 24 = 6 + 24 + \sqrt{6 \times 24} = 30 + \sqrt{144}$
 $= 30 + 12 = 42$.

20. $10y\sqrt{y^3 - y^2} = 10 \times 5 \sqrt{5^3 - 5^2}$
 $= 50 \times \sqrt{125 - 25} = 50 \times \sqrt{100}$
 $= 50 \times 10 = 500$.

21. $\sqrt{110 \frac{1}{4}} = \sqrt{\frac{441}{4}} = \frac{\sqrt{441}}{\sqrt{4}} = \frac{21}{2} = 10.5$.

22. $\sqrt{\frac{25}{81} - \frac{1}{9}} = \sqrt{\frac{25 - 9}{81}} = \sqrt{\frac{16}{81}} = \frac{\sqrt{16}}{\sqrt{81}} = \frac{4}{9}$.

23. Let $[(\sqrt{81})^2]^2 = x^2$. Then, $x^2 = (81)^2$ or $x = 81$.

24.
$$\begin{array}{r} 1 \overline{) 15876} \quad (126 \\ \underline{1} \\ 22 \\ \underline{44} \\ 246 \\ \underline{1476} \\ 1476 \\ \underline{00} \\ 00 \end{array} \quad \therefore \sqrt{15876} = 126$$

25.
$$\begin{array}{r} 1 \overline{) 3.000000} \quad (1.732 \\ \underline{1} \\ 27 \\ \underline{20} \\ 343 \\ \underline{1100} \\ 1029 \\ 3492 \\ \underline{7100} \\ 6984 \\ \underline{116} \end{array} \quad \therefore \sqrt{3} = 1.73 = \frac{173}{100}$$

26. A number ending in 8 can never be a perfect square.

27. The squares of numbers having 1 and 9 as the unit's digit end in the digit 1.

Such numbers are: 1, 9, 11, 19, 21, 29, 31, 39, 41, 49 i.e., there are 10 such numbers.

$$\therefore \text{Required percentage} = \left(\frac{10}{50} \times 100 \right) \% = 20\%.$$

28. Let the given number be x .

$$\text{Then, } x^2 - 25 = (x - 25)^2 \Leftrightarrow x^2 - 25 = x^2 + 625 - 50x \Leftrightarrow 50x = 650 \Leftrightarrow x = 13.$$

29. $(11)^2 = 121$ and $(17)^2 = 289$.

So, the perfect squares between 120 and 300 are the squares of numbers from 11 to 17. Clearly, these are 7 in number.

30. The first perfect square number after 50 is 64 ($= 8^2$) and the last perfect square number before 1000 is 961 ($= (31)^2$). So, the perfect squares between 50 and 1000 are the squares of numbers from 8 to 31. Clearly, these are 24 in number.

31. Clearly, the man was born between 1800 and 1850.

The only perfect square number between 1800 and 1850 is 1849. And, $1849 = (43)^2$.

So, the man was 43 years old in 1849. Thus, he was born in $(1849 - 43) = 1806$.

$$32. Q = \sqrt{\frac{(8R)^2}{4}} = \frac{\sqrt{(8R)^2}}{\sqrt{4}} = \frac{8R}{2} = 4R.$$

33. The smallest such number is 1444 [$= (38)^2$]. It lies between 1000 and 2000.

$$34. \left(\sqrt{2} + \frac{1}{\sqrt{2}} \right)^2 = (\sqrt{2})^2 + \left(\frac{1}{\sqrt{2}} \right)^2 + 2 \times \sqrt{2} \times \frac{1}{\sqrt{2}} \\ = 2 + \frac{1}{2} + 2 = 4\frac{1}{2}.$$

35. We have:

$$1 \times 2 \times 3 \times 4 = 24 \text{ and } 24 + 1 = 25 [= 5^2].$$

$$2 \times 3 \times 4 \times 5 = 120 \text{ and } 120 + 1 = 121 [= 11^2]$$

$$3 \times 4 \times 5 \times 6 = 360 \text{ and } 360 + 1 = 361 [= 19^2]$$

$$4 \times 5 \times 6 \times 7 = 840 \text{ and } 840 + 1 = 841 [= 29^2].$$

$$\therefore p = 1.$$

$$36. \sqrt{0.16} = \frac{\sqrt{16}}{\sqrt{100}} = \frac{4}{10} = 0.4.$$

$$37. \sqrt{0.000441} = \frac{\sqrt{441}}{\sqrt{10^6}} = \frac{\sqrt{441}}{\sqrt{10^6}} = \frac{21}{10^3} = \frac{21}{1000} = 0.021.$$

$$38. \sqrt{0.00004761} = \frac{\sqrt{4761}}{\sqrt{10^8}} = \frac{\sqrt{4761}}{\sqrt{10^8}} = \frac{69}{10^4} = \frac{69}{10000} = 0.0069.$$

$$39. 1.5^2 \times \sqrt{0.0225} = 1.5^2 \times \sqrt{\frac{225}{10000}} = 2.25 \times \frac{15}{100} \\ = 2.25 \times 0.15 = 0.3375.$$

$$40. \sqrt{0.01 + \sqrt{0.0064}} = \sqrt{0.01 + \sqrt{\frac{64}{10000}}} \\ = \sqrt{0.01 + \frac{8}{100}} = \sqrt{0.01 + 0.08} = \sqrt{0.09} = 0.3.$$

$$41. \text{Given exp.} = \sqrt{\frac{1}{100}} + \sqrt{\frac{81}{100}} + \sqrt{\frac{121}{100}} + \sqrt{\frac{9}{10000}} \\ = \frac{1}{10} + \frac{9}{10} + \frac{11}{10} + \frac{3}{100} \\ = 0.1 + 0.9 + 1.1 + 0.03 = 2.13.$$

$$42. \text{Given exp.} = \sqrt{\frac{25}{10000}} \times \sqrt{\frac{225}{100}} \times \sqrt{\frac{1}{10000}} \\ = \frac{5}{100} \times \frac{15}{10} \times \frac{1}{100} = \frac{75}{100000} = 0.00075.$$

1	1.56 25 (1.25
	1
22	56
	44
245	12 25
	12 25
	×

$$\therefore \sqrt{1.5625} = 1.25.$$

$$44. \sqrt{6760000} = \sqrt{0.00000676 \times 10^{12}} = \sqrt{0.00000676} \times \sqrt{10^{12}} \\ = .0026 \times 10^6 = 2600.$$

$$45. \text{Given exp.} = \sqrt{\frac{18225}{10^2}} + \sqrt{\frac{18225}{10^4}} + \sqrt{\frac{18225}{10^6}} + \sqrt{\frac{18225}{10^8}} \\ = \frac{\sqrt{18225}}{10} + \frac{\sqrt{18225}}{10^2} + \frac{\sqrt{18225}}{10^3} + \frac{\sqrt{18225}}{10^4} \\ = \frac{135}{10} + \frac{135}{100} + \frac{135}{1000} + \frac{135}{10000} \\ = 13.5 + 1.35 + 0.135 + 0.0135 = 14.9985.$$

$$46. \text{Given exp.} = \sqrt{\frac{4096}{10^2}} + \sqrt{\frac{4096}{10^4}} + \sqrt{\frac{4096}{10^6}} + \sqrt{\frac{4096}{10^8}} \\ = \frac{\sqrt{4096}}{10} + \frac{\sqrt{4096}}{10^2} + \frac{\sqrt{4096}}{10^3} + \frac{\sqrt{4096}}{10^4} \\ = \frac{64}{10} + \frac{64}{100} + \frac{64}{1000} + \frac{64}{10000} \\ = 6.4 + 0.64 + 0.064 + 0.0064 \\ = 7.1104 \approx 7.11.$$

$$47. \text{Given exp.} = \sqrt{1.30} + \sqrt{1300} + \sqrt{0.0130} \\ = \sqrt{\frac{130}{100}} + \sqrt{13 \times 100} + \sqrt{\frac{130}{10000}} \\ = \frac{\sqrt{130}}{10} + \sqrt{13} \times 10 + \frac{\sqrt{130}}{100} \\ = \frac{11.40}{10} + 3.605 \times 10 + \frac{11.40}{100} \\ = 1.14 + 36.05 + 0.114 = 37.304.$$

$$48. \frac{52}{x} = \sqrt{\frac{169}{289}} \\ \Leftrightarrow \frac{52}{x} = \frac{13}{17} \\ \Leftrightarrow x = \left(\frac{52 \times 17}{13} \right) = 68.$$

49. Let the missing number be x .

$$\text{Then, } x^2 = 15 \times 135 \Leftrightarrow x = \sqrt{15 \times 135} \\ = \sqrt{15^2 \times 3^2} = 15 \times 3 = 45.$$

$$\begin{aligned} 50. \text{ Let } \frac{x}{1776} &= \frac{111}{x}. \text{ Then, } x^2 \\ &= 111 \times 1776 = 111 \times 111 \times 16 \\ \Rightarrow x &= \sqrt{(111)^2 \times (4)^2} = 111 \times 4 = 444. \end{aligned}$$

$$51. \text{ Let } \frac{4\frac{1}{2}}{x} = \frac{x}{32}. \text{ Then, } x^2 = 32 \times \frac{9}{2} = 144 \Leftrightarrow x = \sqrt{144} = 12.$$

$$\begin{aligned} 52. \text{ Let } \frac{x}{\sqrt{128}} &= \frac{\sqrt{162}}{x}. \text{ Then, } x^2 = \sqrt{128 \times 162} = \sqrt{64 \times 2 \times 18 \times 9} \\ &= \sqrt{8^2 \times 6^2 \times 3^2} = 8 \times 6 \times 3 = 144. \\ \therefore x &= \sqrt{144} = 12. \end{aligned}$$

$$\begin{aligned} 53. \sqrt{x + \frac{x}{y}} &= x \sqrt{\frac{x}{y}} \Rightarrow x + \frac{x}{y} = x^2 \cdot \frac{x}{y} \Rightarrow \frac{xy + x}{y} = \frac{x^3}{y} \\ \Rightarrow xy + x &= x^3 \\ \Rightarrow y + 1 &= x^2 \\ \Rightarrow y &= x^2 - 1. \end{aligned}$$

$$\begin{aligned} 54. n^2 &= (25)^{64} \times (64)^{25} = (5^2)^{64} \times (2^6)^{25} \\ &= 5^{128} \times 2^{150} = 5^{128} \times 2^{128} \times 2^{22} \\ \Rightarrow n &= 5^{64} \times 2^{64} \times 2^{11} = (5 \times 2)^{64} \times 2^{11} = 10^{64} \times 2048. \\ \therefore \text{ Sum of digits of } n &= 2 + 0 + 4 + 8 = 14. \end{aligned}$$

$$\begin{aligned} 55. \frac{0.13}{p^2} &= 13 \\ \Leftrightarrow p^2 &= \frac{0.13}{13} = \frac{1}{100} \\ \Leftrightarrow p &= \sqrt{\frac{1}{100}} = \frac{1}{10} = 0.1. \end{aligned}$$

$$56. \text{ Let the required number be } x. \text{ Then, } \frac{x}{\sqrt{0.25}} = 25$$

$$\begin{aligned} \Leftrightarrow \frac{x}{0.5} &= 25 \\ \Leftrightarrow x &= 25 \times 0.5 = 12.5. \end{aligned}$$

$$57. \sqrt{3^n} = 729 = 3^6 \Leftrightarrow (\sqrt{3^n})^2 = (3^6)^2 \Leftrightarrow 3^n = 3^{12} \Leftrightarrow n = 12.$$

$$\begin{aligned} 58. \sqrt{18 \times 14 \times x} &= 84 \Leftrightarrow 18 \times 14 \times x = 84 \times 84 \\ \Leftrightarrow x &= \frac{84 \times 84}{18 \times 14} = 28. \end{aligned}$$

$$\begin{aligned} 59. \text{ Let } 28\sqrt{x} + 1426 &= 3 \times 718. \\ \text{Then, } 28\sqrt{x} &= 2154 - 1426 \Leftrightarrow 28\sqrt{x} = 728 \Leftrightarrow \sqrt{x} = 26 \\ \Leftrightarrow x &= (26)^2 = 676. \end{aligned}$$

$$\begin{aligned} 60. \text{ Let } \sqrt{\frac{x}{169}} &= \frac{54}{39}. \text{ Then, } \frac{\sqrt{x}}{13} = \frac{54}{39} \Leftrightarrow \sqrt{x} = \left(\frac{54}{39} \times 13\right) = 18 \\ \Leftrightarrow x &= (18)^2 = 324. \end{aligned}$$

$$\begin{aligned} 61. \frac{\sqrt{x}}{\sqrt{441}} &= 0.02 \Leftrightarrow \frac{\sqrt{x}}{21} = 0.02 \\ \Leftrightarrow \sqrt{x} &= 0.02 \times 21 = 0.42 \Leftrightarrow x \\ &= (0.42)^2 = 0.1764. \end{aligned}$$

$$\begin{aligned} 62. \text{ Let } \sqrt{\frac{.0196}{x}} &= 0.2. \text{ Then, } \frac{.0196}{x} = 0.04 \\ \Leftrightarrow x &= \frac{.0196}{.04} = \frac{1.96}{4} = .49. \end{aligned}$$

$$\begin{aligned} 63. \text{ Let } \sqrt{0.0169 \times x} &= 1.3. \text{ Then, } 0.0169x = (1.3)^2 = 1.69 \\ \Rightarrow x &= \frac{1.69}{0.0169} = 100. \end{aligned}$$

$$\begin{aligned} 64. 37 + \sqrt{.0615 + x} &= 37.25 \Leftrightarrow \sqrt{.0615 + x} = 0.25 \\ \Leftrightarrow .0615 + x &= (0.25)^2 = 0.0625 \\ \Leftrightarrow x &= .001 = \frac{1}{10^3} = 10^{-3}. \end{aligned}$$

$$\begin{aligned} 65. \sqrt{(x-1)(y+2)} &= 7 \Rightarrow (x-1)(y+2) = (7)^2 \Rightarrow (x-1) = 7 \\ \text{and } (y+2) &= 7 \Rightarrow x = 8 \text{ and } y = 5. \end{aligned}$$

$$\begin{aligned} 66. \frac{\sqrt{a}}{\sqrt{b}} &= \frac{.004 \times .4}{\sqrt{.04 \times .4}} \Rightarrow \frac{a}{b} = \frac{.004 \times .4 \times .004 \times .4}{.04 \times .4} = \frac{.0000064}{.04} \\ \Rightarrow \frac{a}{b} &= \frac{.00064}{4} = .00016 = \frac{16}{10^5} = 16 \times 10^{-5}. \end{aligned}$$

$$\begin{aligned} 67. \text{ Let the number be } x. \text{ Then, } \\ \frac{3}{5}x^2 &= 126.15 \Leftrightarrow x^2 = \left(126.15 \times \frac{5}{3}\right) = 210.25 \\ \Leftrightarrow x &= \sqrt{210.25} = 14.5. \end{aligned}$$

$$68. \sqrt{\frac{0.361}{0.00169}} = \sqrt{\frac{0.36100}{0.00169}} = \sqrt{\frac{36100}{169}} = \frac{190}{13}.$$

$$69. \sqrt{\frac{48.4}{0.289}} = \sqrt{\frac{48.400}{0.289}} = \sqrt{\frac{48400}{289}} = \frac{220}{17} = 12\frac{16}{17}.$$

$$\begin{aligned} 70. \sqrt{1 + \frac{x}{169}} &= \frac{14}{13} \Rightarrow 1 + \frac{x}{169} = \frac{196}{169} \\ \Rightarrow \frac{x}{169} &= \left(\frac{196}{169} - 1\right) = \frac{27}{169} \Rightarrow x = 27. \end{aligned}$$

$$\begin{aligned} 71. \sqrt{1 + \frac{55}{729}} &= 1 + \frac{x}{27} \Rightarrow \sqrt{\frac{784}{729}} = \frac{27+x}{27} \\ \Rightarrow \frac{28}{27} &= \frac{27+x}{27} \Rightarrow 27+x = 28 \Rightarrow x = 1. \end{aligned}$$

$$72. \frac{\sqrt{4}}{\sqrt{3}} - \frac{\sqrt{3}}{\sqrt{4}} = \frac{\sqrt{4}}{\sqrt{3}} - \frac{\sqrt{3}}{\sqrt{4}} = \frac{\sqrt{4} \times \sqrt{4} - \sqrt{3} \times \sqrt{3}}{\sqrt{12}} = \frac{4-3}{2\sqrt{3}} = \frac{1}{2\sqrt{3}}.$$

$$\begin{aligned} 73. \quad & \begin{array}{r} 1 \overline{) 2.00 \ 00 \ 00} \ (1.414 \\ \underline{1} \\ 100 \\ \underline{84} \\ 160 \\ \underline{128} \\ 320 \\ \underline{288} \\ 320 \\ \underline{288} \\ 320 \\ \underline{288} \\ 320 \end{array} \\ & \therefore \sqrt{2} = 1.414 \end{aligned}$$

$$\begin{aligned} 74. 2\sqrt{27} - \sqrt{75} + \sqrt{12} &= 2\sqrt{9 \times 3} - \sqrt{25 \times 3} + \sqrt{4 \times 3} \\ &= 6\sqrt{3} - 5\sqrt{3} + 2\sqrt{3} = 3\sqrt{3}. \end{aligned}$$

$$\begin{aligned}
 75. (\sqrt{12} + \sqrt{18}) - (\sqrt{3} + \sqrt{2}) &= (\sqrt{4 \times 3} + \sqrt{9 \times 2}) - (\sqrt{3} + \sqrt{2}) \\
 &= (2\sqrt{3} + 3\sqrt{2}) - (\sqrt{3} + \sqrt{2}) \\
 &= (2\sqrt{3} - \sqrt{3}) + (3\sqrt{2} - \sqrt{2}) = \sqrt{3} + 2\sqrt{2}.
 \end{aligned}$$

$$76. \frac{\sqrt{24} + \sqrt{216}}{\sqrt{96}} = \frac{\sqrt{4 \times 6} + \sqrt{36 \times 6}}{\sqrt{16 \times 6}} = \frac{2\sqrt{6} + 6\sqrt{6}}{4\sqrt{6}} = \frac{8\sqrt{6}}{4\sqrt{6}} = 2.$$

$$\begin{aligned}
 77. \frac{\sqrt{80} - \sqrt{112}}{\sqrt{45} - \sqrt{63}} &= \frac{\sqrt{16 \times 5} - \sqrt{16 \times 7}}{\sqrt{9 \times 5} - \sqrt{9 \times 7}} \\
 &= \frac{4\sqrt{5} - 4\sqrt{7}}{3\sqrt{5} - 3\sqrt{7}} = \frac{4(\sqrt{5} - \sqrt{7})}{3(\sqrt{5} - \sqrt{7})} = \frac{4}{3} = 1\frac{1}{3}.
 \end{aligned}$$

$$\begin{aligned}
 78. 3\sqrt{5} + \sqrt{125} &= 17.88 \Rightarrow 3\sqrt{5} + \sqrt{25 \times 5} = 17.88 \\
 \Rightarrow 3\sqrt{5} + 5\sqrt{5} &= 17.88 \Rightarrow 8\sqrt{5} = 17.88 \Rightarrow \sqrt{5} = 2.235. \\
 \therefore \sqrt{80} + 6\sqrt{5} &= \sqrt{16 \times 5} + 6\sqrt{5} = 4\sqrt{5} + 6\sqrt{5} \\
 &= 10\sqrt{5} = (10 \times 2.235) = 22.35.
 \end{aligned}$$

$$79. \sqrt{50} \times \sqrt{98} = \sqrt{50 \times 98} = \sqrt{4900} = 70.$$

$$\begin{aligned}
 80. \text{Given exp.} &= \sqrt{4 \times 2} + 2\sqrt{16 \times 2} - 3\sqrt{64 \times 2} + 4\sqrt{25 \times 2} \\
 &= 2\sqrt{2} + 8\sqrt{2} - 24\sqrt{2} + 20\sqrt{2} = 6\sqrt{2} = 6 \times 1.414 = 8.484.
 \end{aligned}$$

$$\begin{aligned}
 81. \text{Given exp.} &= \frac{3\sqrt{12}}{2\sqrt{28}} \times \frac{\sqrt{98}}{2\sqrt{21}} = \frac{3\sqrt{4 \times 3}}{2\sqrt{4 \times 7}} \times \frac{\sqrt{49 \times 2}}{2\sqrt{21}} \\
 &= \frac{6\sqrt{3}}{4\sqrt{7}} \times \frac{7\sqrt{2}}{2\sqrt{21}} = \frac{21\sqrt{6}}{4\sqrt{7 \times 21}} = \frac{21\sqrt{6}}{28\sqrt{3}} \\
 &= \frac{3}{4}\sqrt{2} = \frac{3}{4} \times 1.414 = 3 \times 0.3535 = 1.0605.
 \end{aligned}$$

$$\begin{aligned}
 82. \text{Given exp.} &= \sqrt{\frac{11025}{100}} \times \sqrt{\frac{1}{100}} + \sqrt{\frac{25}{10000}} - \sqrt{\frac{42025}{100}} \\
 &= \frac{105}{10} \times \frac{1}{10} + \frac{5}{100} - \frac{205}{10} \\
 &= \frac{105}{100} \times \frac{100}{5} - \frac{205}{10} \\
 &= 21 - \frac{205}{10} = \frac{5}{10} = \frac{1}{2} = 0.50.
 \end{aligned}$$

83. Sum of decimal places in the numerator and denominator under the radical sign being the same, we remove the decimal.

$$\therefore \text{Given exp.} = \sqrt{\frac{81 \times 484}{64 \times 625}} = \frac{9 \times 22}{8 \times 25} = 0.99.$$

$$84. \text{Given exp.} = \sqrt{\frac{204 \times 42}{7 \times 34}} = \sqrt{36} = 6.$$

$$\begin{aligned}
 85. \text{Given exp.} &= \sqrt{\frac{81 \times 324 \times 4624}{15625 \times 289 \times 729 \times 64}} \\
 &= \frac{9 \times 18 \times 68}{125 \times 17 \times 27 \times 8} = \frac{3}{125} = 0.024.
 \end{aligned}$$

$$\begin{aligned}
 86. \text{Given exp.} &= \sqrt{\frac{9.5 \times .08500}{.19 \times .0017}} = \sqrt{\frac{95 \times 8500}{19 \times 17}} \\
 &= \sqrt{5 \times 500} = \sqrt{2500} = 50.
 \end{aligned}$$

$$\begin{aligned}
 87. \text{Given exp.} &= \sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{\left(\frac{0.03}{10}\right)^2 + \left(\frac{0.21}{10}\right)^2 + \left(\frac{0.065}{10}\right)^2}} \\
 &= \sqrt{\frac{100[(0.03)^2 + (0.21)^2 + (0.065)^2]}{(0.03)^2 + (0.21)^2 + (0.065)^2}} = \sqrt{100} = 10.
 \end{aligned}$$

$$\begin{aligned}
 88. \sqrt{(7 + 3\sqrt{5})(7 - 3\sqrt{5})} &= \sqrt{(7)^2 - (3\sqrt{5})^2} \\
 &= \sqrt{49 - 45} = \sqrt{4} = 2.
 \end{aligned}$$

$$\begin{aligned}
 89. \left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)^2 &= (\sqrt{3})^2 + \left(\frac{1}{\sqrt{3}}\right)^2 - 2 \times \sqrt{3} \times \frac{1}{\sqrt{3}} \\
 &= 3 + \frac{1}{3} - 2 = 1 + \frac{1}{3} = \frac{4}{3}.
 \end{aligned}$$

$$\begin{aligned}
 90. \sqrt{4a^2 - 4a + 1} + 3a \\
 &= \sqrt{(1)^2 + (2a)^2 - 2 \times 1 \times 2a} + 3a = \sqrt{(1 - 2a)^2} + 3a \\
 &= (1 - 2a) + 3a = (1 + a) = (1 + 0.1039) = 1.1039.
 \end{aligned}$$

$$\begin{aligned}
 91. \sqrt{\frac{(0.75)^3}{(1 - 0.75)} + [0.75 + (0.75)^2 + 1]} \\
 &= \sqrt{\frac{(0.75)^3 + (1 - 0.75)[(1)^2 + (0.75)^2 + 1 \times 0.75]}{1 - 0.75}} \\
 &= \sqrt{\frac{(0.75)^3 + [(1)^3 - (0.75)^3]}{1 - 0.75}} \\
 &= \sqrt{\frac{1}{0.25}} = \sqrt{\frac{100}{25}} = \sqrt{4} = 2.
 \end{aligned}$$

$$\begin{aligned}
 92. a + b + c &= 27\sqrt{29} \Rightarrow 2c + \frac{3}{2}c + c = 27\sqrt{29} \\
 \Rightarrow \frac{9}{2}c &= 27\sqrt{29} \Rightarrow c = 6\sqrt{29}.
 \end{aligned}$$

$$\begin{aligned}
 \sqrt{a^2 + b^2 + c^2} &= \sqrt{(a + b + c)^2 - 2(ab + bc + ca)} \\
 &= \sqrt{(27\sqrt{29})^2 - 2\left(2c \times \frac{3}{2}c + \frac{3}{2}c \times c + c \times 2c\right)} \\
 &= \sqrt{(729 \times 29) - 2\left(3c^2 + \frac{3}{2}c^2 + 2c^2\right)} \\
 &= \sqrt{(729 \times 29) - 2 \times \frac{13}{2}c^2} \\
 &= \sqrt{(729 \times 29) - 13 \times (6\sqrt{29})^2} = \sqrt{29(729 - 468)} \\
 &= \sqrt{29 \times 261} = \sqrt{29 \times 29 \times 9} = 29 \times 3 = 87.
 \end{aligned}$$

$$93. \sqrt{0.4} = \sqrt{\frac{4}{10}} = \frac{2}{\sqrt{10}} = 0.666..... = 0.\bar{6}.$$

$$94. \sqrt{0.09} = \sqrt{\frac{9}{100}} = \frac{3}{10} = 0.3, \text{ which is rational.}$$

95.

$$\begin{array}{r} 6 \overline{) 0.40 \ 00 \ 00} \ (.63 \\ \underline{36} \\ 4 \ 00 \\ \underline{3 \ 69} \end{array}$$

$$\therefore \sqrt{0.4} = 0.63$$

96.

$$\begin{array}{r} 4 \overline{) 0.20 \ 00 \ 00} \ (.447 \\ \underline{16} \\ 4 \ 00 \\ \underline{3 \ 36} \\ 887 \\ \underline{64 \ 00} \\ 62 \ 09 \end{array}$$

$$\therefore \sqrt{0.2} = 0.447$$

97.

$$\begin{array}{r} 3 \overline{) 0.12 \ 10 \ 00} \ (.347 \\ \underline{9} \\ 3 \ 10 \\ \underline{2 \ 56} \\ 687 \\ \underline{54 \ 00} \\ 48 \ 00 \end{array}$$

$$\therefore \sqrt{0.121} = 0.347$$

98.

$$\begin{array}{r} 2 \overline{) 0.06 \ 40 \ 00} \ (.252 \\ \underline{4} \\ 2 \ 40 \\ \underline{2 \ 25} \\ 502 \\ \underline{1500} \\ 1004 \end{array}$$

$$\therefore \sqrt{0.064} = 0.252$$

99. $\sqrt{\frac{0.16}{0.4}} = \sqrt{\frac{0.16}{0.40}} = \sqrt{\frac{16}{40}} = \sqrt{\frac{4}{10}} = \sqrt{0.4} = 0.63.$

100. $\frac{1 + \sqrt{0.01}}{1 - \sqrt{0.1}} = \frac{1 + 0.1}{1 - 0.316} = \frac{1.1}{0.684} = \frac{1100}{684} = 1.6.$

$$\begin{array}{r} 3 \overline{) 0.10 \ 00 \ 00} \ (.0316 \\ \underline{9} \\ 61 \\ \underline{1 \ 00} \\ 61 \\ \underline{39 \ 00} \\ 37 \ 56 \end{array}$$

101. $\sqrt{535.9225} = \sqrt{\frac{5359225}{10000}} = \frac{2315}{100} = 23.15.$

$$\begin{array}{r} 2 \overline{) 5 \ 35 \ 92 \ 25} \ (2315 \\ \underline{4} \\ 43 \\ \underline{1 \ 35} \\ 1 \ 29 \\ \underline{461} \\ 461 \\ \underline{4 \ 61} \\ 4625 \\ \underline{231 \ 25} \\ 231 \ 25 \\ \underline{ \times} \end{array}$$

102. $\frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5} = \frac{2.236}{5} = 0.447.$

103. $\sqrt{\frac{8}{3}} = \sqrt{\frac{8 \times 3}{3 \times 3}} = \frac{\sqrt{24}}{3} = \frac{4.899}{3} = 1.633.$

104. $\frac{3\sqrt{2}}{2\sqrt{3}} = \frac{3\sqrt{2}}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{3\sqrt{6}}{2 \times 3} = \frac{\sqrt{6}}{2} = \frac{2.449}{2} = 1.2245.$

105. $\frac{\sqrt{5}}{2} - \frac{10}{\sqrt{5}} + \sqrt{125} = \frac{(\sqrt{5})^2 - 20 + 2\sqrt{5} \times 5\sqrt{5}}{2\sqrt{5}}$
 $= \frac{5 - 20 + 50}{2\sqrt{5}} = \frac{35}{2\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{35\sqrt{5}}{10} = \frac{7}{2} \times 2.236$
 $= 7 \times 1.118 = 7.826.$

106. Clearly, $a * b = \sqrt{a^2 + b^2}$. $\therefore 5 * 12$

$$= \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

107. Let the missing digit be x .

$$\begin{array}{r} 1 \overline{) 1 \ 53 \ 7x} \ (124 \\ \underline{1} \\ 22 \\ \underline{53} \\ 44 \\ \underline{244} \\ 9 \ 7x \\ \underline{9 \ 76} \\ \times \end{array}$$

Then, $x = 6$.

108. $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$
 $= 7 \times 2 \times 3 \times 5 \times 2^2 \times 3 \times 2$
 $= 2^4 \times 3^2 \times 5 \times 7.$

Thus, the smallest perfect square number which is divisible by $7!$ is

$$(2^4 \times 3^2 \times 5 \times 7) \times (5 \times 7)$$

 $= 5040 \times 35 = 176400.$

109. L.C.M. of 3, 4, 5, 6, 8 is 120. Now, 120
 $= 2 \times 2 \times 2 \times 3 \times 5.$

To make it a perfect square, it must be multiplied by $2 \times 3 \times 5$.

$$\text{So, required number} = 2^2 \times 2^2 \times 3^2 \times 5^2 = 3600.$$

110. L.C.M. of 21, 36, 66

$$= 2772. \text{ Now, } 2772$$

 $= 2 \times 2 \times 3 \times 3 \times 7 \times 11.$

To make it a perfect square, it must be multiplied by 7×11 .

$$\text{So, required number} = 2^2 \times 3^2 \times 7^2 \times 11^2 = 213444.$$

111. $294 = 7 \times 7 \times 2 \times 3.$

To make it a perfect square, it must be multiplied by 2×3 i.e., 6.

$$\therefore \text{Required number} = 6.$$

112. $5808 = 2 \times 2 \times 2 \times 2 \times 3 \times 11 \times 11 = 2^2 \times 2^2 \times 3 \times 11^2.$

To make it a perfect square, it must be multiplied by 3.

113. $1470 = 7 \times 7 \times 5 \times 6.$ To make it a perfect square, it must be divided by 5×6 , i.e., 30.

114.

$$\begin{array}{r}
 7 \overline{) 54 \ 91 \ 62 \ (741)} \\
 \underline{49} \\
 144 \\
 \underline{5 \ 91} \\
 1481 \\
 \underline{15 \ 62} \\
 \underline{14 \ 81} \\
 81
 \end{array}$$

∴ Required number to be subtracted = 81.

115. $0.000326 = \frac{326}{10^6}$

$$\begin{array}{r}
 1 \overline{) 3 \ 26 \ (18)} \\
 \underline{1} \\
 28 \\
 \underline{2 \ 26} \\
 2 \ 24 \\
 \underline{2} \\
 2
 \end{array}$$

∴ Required number to be subtracted = $\frac{2}{10^6} = 0.000002$.

116.

$$\begin{array}{r}
 8 \overline{) 77 \ 00 \ (87)} \\
 \underline{64} \\
 167 \\
 \underline{13 \ 00} \\
 11 \ 69 \\
 \underline{1 \ 31} \\
 1
 \end{array}$$

∴ Number to be added
 $= (88)^2 - 7700 = 7744 - 7700 = 44$.

117.

$$\begin{array}{r}
 8 \overline{) 68 \ 06 \ 21 \ (824)} \\
 \underline{64} \\
 162 \\
 \underline{4 \ 06} \\
 3 \ 24 \\
 \underline{3 \ 24} \\
 1644 \\
 \underline{82 \ 21} \\
 65 \ 76 \\
 \underline{16 \ 45} \\
 11
 \end{array}$$

∴ Number to be added = $(825)^2 - 680621$
 $= 680625 - 680621 = 4$.

118. Greatest number of four digits is 9999.

$$\begin{array}{r}
 9 \overline{) 99 \ 99 \ (99)} \\
 \underline{81} \\
 189 \\
 \underline{17 \ 01} \\
 1 \ 98
 \end{array}$$

∴ Reqd. number = $(9999 - 198) = 9801$.

119. Least number of 4 digits is 1000.

$$\begin{array}{r}
 3 \overline{) 10 \ 00 \ (31)} \\
 \underline{9} \\
 61 \\
 \underline{1 \ 00} \\
 61 \\
 \underline{39} \\
 22
 \end{array}$$

∴ $(31)^2 < 1000 < (32)^2$.

Hence, required number = $(32)^2 = 1024$.

120. Let the 18 consecutive natural numbers be $x, (x+1), (x+2), (x+3), \dots, (x+17)$. Then, $x + (x+1) + (x+2) + \dots + (x+17) = 18x + (1+2+3+\dots+17) = 18x + 153$.

Putting $x = 1, 2, 3, 4, \dots$ we find that the smallest value of x for which $(18x + 153)$ becomes a perfect square is $x = 4$.

∴ Required value = $18 \times 4 + 153 = 72 + 153 = 225$.

121. Given expression

$$\begin{aligned}
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2^2 - (\sqrt{2+\sqrt{2+\sqrt{3}}})^2} \\
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{4 - (2+\sqrt{2+\sqrt{3}})} \\
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2-\sqrt{2+\sqrt{3}}} \\
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2^2 - (\sqrt{2+\sqrt{3}})^2} = \sqrt{2+\sqrt{3}} \cdot \sqrt{2-\sqrt{3}} \\
 &= \sqrt{2^2 - (\sqrt{3})^2} = \sqrt{4-3} = \sqrt{1} = 1.
 \end{aligned}$$

$$\begin{aligned}
 122. \frac{1}{(\sqrt{5}-\sqrt{3})} &= \frac{1}{(\sqrt{5}-\sqrt{3})} \times \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})} = \frac{(\sqrt{5}+\sqrt{3})}{(5-3)} \\
 &= \frac{(2.2361+1.7321)}{2} = \frac{3.9682}{2} = 1.9841.
 \end{aligned}$$

$$\begin{aligned}
 123. \text{ Given expression } &= \frac{1}{(\sqrt{9}-\sqrt{8})} \times \frac{(\sqrt{9}+\sqrt{8})}{(\sqrt{9}+\sqrt{8})} - \frac{1}{(\sqrt{8}-\sqrt{7})} \\
 &\quad \times \frac{(\sqrt{8}+\sqrt{7})}{(\sqrt{8}+\sqrt{7})} + \frac{1}{(\sqrt{7}-\sqrt{6})} \times \frac{(\sqrt{7}+\sqrt{6})}{(\sqrt{7}+\sqrt{6})} \\
 &\quad - \frac{1}{(\sqrt{6}-\sqrt{5})} \times \frac{(\sqrt{6}+\sqrt{5})}{(\sqrt{6}+\sqrt{5})} + \frac{1}{(\sqrt{5}-\sqrt{4})} \times \frac{(\sqrt{5}+\sqrt{4})}{(\sqrt{5}+\sqrt{4})} \\
 &= \frac{(\sqrt{9}+\sqrt{8})}{(9-8)} - \frac{(\sqrt{8}+\sqrt{7})}{(8-7)} + \frac{(\sqrt{7}+\sqrt{6})}{(7-6)} \\
 &\quad - \frac{(\sqrt{6}+\sqrt{5})}{(6-5)} + \frac{(\sqrt{5}+\sqrt{4})}{(5-4)} \\
 &= (\sqrt{9}+\sqrt{8}) - (\sqrt{8}+\sqrt{7}) + (\sqrt{7}+\sqrt{6}) - (\sqrt{6}+\sqrt{5}) \\
 &\quad + (\sqrt{5}+\sqrt{4}) = (\sqrt{9}+\sqrt{4}) = 3+2=5.
 \end{aligned}$$

$$\begin{aligned}
 124. \text{ Given exp. } &= \frac{1}{\sqrt{2}+\sqrt{1}} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{\sqrt{4}+\sqrt{3}} \\
 &\quad + \dots + \frac{1}{\sqrt{121}+\sqrt{120}} \\
 &= \frac{1}{\sqrt{2}+\sqrt{1}} \times \frac{\sqrt{2}-\sqrt{1}}{\sqrt{2}-\sqrt{1}} + \frac{1}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}} \\
 &\quad + \frac{1}{\sqrt{4}+\sqrt{3}} \times \frac{\sqrt{4}-\sqrt{3}}{\sqrt{4}-\sqrt{3}} + \dots + \frac{1}{\sqrt{121}+\sqrt{120}} \times \frac{\sqrt{121}-\sqrt{120}}{\sqrt{121}-\sqrt{120}} \\
 &= \frac{\sqrt{2}-\sqrt{1}}{2-1} + \frac{\sqrt{3}-\sqrt{2}}{3-2} + \frac{\sqrt{4}-\sqrt{3}}{4-3} + \dots + \frac{\sqrt{121}-\sqrt{120}}{121-120} \\
 &= \sqrt{2}-\sqrt{1}+\sqrt{3}-\sqrt{2}+\sqrt{4}-\sqrt{3}+\dots+\sqrt{121}-\sqrt{120} \\
 &= -1+\sqrt{121} = -1+11=10.
 \end{aligned}$$

$$\begin{aligned}
 125. \text{ Given exp. } &= \frac{(1+\sqrt{3})(1-\sqrt{3}) - (1-\sqrt{3}) + (1+\sqrt{3})}{(1+\sqrt{3})(1-\sqrt{3})} \\
 &= \frac{1 - (\sqrt{3})^2 - 1 + \sqrt{3} + 1 + \sqrt{3}}{1^2 - (\sqrt{3})^2} \\
 &= \frac{2\sqrt{3} - 2}{1 - 3} = \frac{2(\sqrt{3} - 1)}{(-2)} = 1 - \sqrt{3}.
 \end{aligned}$$

$$\begin{aligned}
 126. \text{ Given exp. } &= (2+\sqrt{2}) + \frac{1}{(2+\sqrt{2})} \times \frac{(2-\sqrt{2})}{(2-\sqrt{2})} \\
 &\quad - \frac{1}{(2-\sqrt{2})} \times \frac{(2+\sqrt{2})}{(2+\sqrt{2})} \\
 &= (2+\sqrt{2}) + \frac{(2-\sqrt{2})}{(4-2)} - \frac{(2+\sqrt{2})}{(4-2)} \\
 &= (2+\sqrt{2}) + \frac{1}{2}(2-\sqrt{2}) - \frac{1}{2}(2+\sqrt{2}) = 2.
 \end{aligned}$$

$$\begin{aligned}
 127. \text{ Given exp. } &= \frac{1}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})} + \frac{2}{(3+\sqrt{5})} \\
 &\quad \times \frac{(3-\sqrt{5})}{(3-\sqrt{5})} - \frac{3}{(3+\sqrt{3})} \times \frac{(3-\sqrt{3})}{(3-\sqrt{3})} \\
 &= \frac{(\sqrt{5}-\sqrt{3})}{(5-3)} + \frac{2(3-\sqrt{5})}{(9-5)} - \frac{3(3-\sqrt{3})}{(9-3)} \\
 &= \frac{\sqrt{5}-\sqrt{3}}{2} + \frac{2(3-\sqrt{5})}{4} - \frac{3(3-\sqrt{3})}{6} \\
 &= \frac{6(\sqrt{5}-\sqrt{3}) + 6(3-\sqrt{5}) - 6(3-\sqrt{3})}{12} = 0.
 \end{aligned}$$

$$\begin{aligned}
 128. \frac{7}{(3+\sqrt{2})} &= \frac{7}{(3+\sqrt{2})} \times \frac{(3-\sqrt{2})}{(3-\sqrt{2})} = \frac{7(3-\sqrt{2})}{(9-2)} \\
 &= (3-\sqrt{2}) = (3-1.4142) = 1.5858.
 \end{aligned}$$

$$\begin{aligned}
 129. \text{ Given exp. } &= \frac{3\sqrt{2}}{(\sqrt{6}-\sqrt{3})} \times \frac{(\sqrt{6}+\sqrt{3})}{(\sqrt{6}+\sqrt{3})} - \frac{4\sqrt{3}}{(\sqrt{6}-\sqrt{2})} \\
 &\quad \times \frac{(\sqrt{6}+\sqrt{2})}{(\sqrt{6}+\sqrt{2})} - \frac{6}{2(\sqrt{2}-\sqrt{3})} \\
 &= \frac{3\sqrt{2}(\sqrt{6}+\sqrt{3})}{(6-3)} - \frac{4\sqrt{3}(\sqrt{6}+\sqrt{2})}{(6-2)} \\
 &\quad + \frac{3}{(\sqrt{3}-\sqrt{2})} \times \frac{(\sqrt{3}+\sqrt{2})}{(\sqrt{3}+\sqrt{2})} \\
 &= \sqrt{2}(\sqrt{6}+\sqrt{3}) - \sqrt{3}(\sqrt{6}+\sqrt{2}) + 3(\sqrt{3}+\sqrt{2}) \\
 &= \sqrt{12} + \sqrt{6} - \sqrt{18} - \sqrt{6} + 3\sqrt{3} + 3\sqrt{2} \\
 &= 2\sqrt{3} - 3\sqrt{2} + 3\sqrt{3} + 3\sqrt{2} = 5\sqrt{3}
 \end{aligned}$$

$$\begin{aligned}
 130. \frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} &= \frac{(\sqrt{7}+\sqrt{5})}{(\sqrt{7}-\sqrt{5})} \times \frac{(\sqrt{7}+\sqrt{5})}{(\sqrt{7}+\sqrt{5})} = \frac{(\sqrt{7}+\sqrt{5})^2}{(7-5)} \\
 &= \frac{7+5+2\sqrt{35}}{2} = \frac{12+2\sqrt{35}}{2} = 6+\sqrt{35}.
 \end{aligned}$$

$$131. a+b\sqrt{3} = \frac{(5+2\sqrt{3})}{(7+4\sqrt{3})} \times \frac{(7-4\sqrt{3})}{(7-4\sqrt{3})} = \frac{35-20\sqrt{3}+14\sqrt{3}-24}{(7)^2-(4\sqrt{3})^2}$$

$$= \frac{11-6\sqrt{3}}{49-48} = 11-6\sqrt{3}. \quad \therefore a = 11, b = -6.$$

$$132. \frac{\sqrt{2}-1}{\sqrt{2}+1} = \frac{(\sqrt{2}-1)}{(\sqrt{2}+1)} \times \frac{(\sqrt{2}-1)}{(\sqrt{2}-1)} = (\sqrt{2}-1)^2.$$

$$\therefore \sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}} = (\sqrt{2}-1) = (1.414-1) = 0.414.$$

$$\begin{aligned}
 133. \text{ Given exp. } &= \frac{3+\sqrt{6}}{5\sqrt{3}-4\sqrt{3}-4\sqrt{2}+5\sqrt{2}} = \frac{(3+\sqrt{6})}{(\sqrt{3}+\sqrt{2})} \\
 &= \frac{(3+\sqrt{6})}{(\sqrt{3}+\sqrt{2})} \times \frac{(\sqrt{3}-\sqrt{2})}{(\sqrt{3}-\sqrt{2})} \\
 &= \frac{3\sqrt{3}-3\sqrt{2}+3\sqrt{2}-2\sqrt{3}}{(3-2)} = \sqrt{3} = 1.732.
 \end{aligned}$$

$$\begin{aligned}
 134. \text{ Given exp. } &= \frac{(2+\sqrt{3})}{(2-\sqrt{3})} \times \frac{(2+\sqrt{3})}{(2+\sqrt{3})} + \frac{(2-\sqrt{3})}{(2+\sqrt{3})} \times \frac{(2-\sqrt{3})}{(2-\sqrt{3})} \\
 &\quad + \frac{(\sqrt{3}-1)}{(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)} \\
 &= \frac{(2+\sqrt{3})^2}{(4-3)} + \frac{(2-\sqrt{3})^2}{(4-3)} + \frac{(\sqrt{3}-1)^2}{(3-1)} \\
 &= [(2+\sqrt{3})^2 + (2-\sqrt{3})^2] + \frac{4-2\sqrt{3}}{2} \\
 &= 2(4+3) + 2 - \sqrt{3} = 16 - \sqrt{3}.
 \end{aligned}$$

$$\begin{aligned}
 135. x+y &= \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} + \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}} = \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}-\sqrt{3})} \times \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \\
 &\quad + \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})} \\
 &= \frac{(\sqrt{5}+\sqrt{3})^2}{(\sqrt{5})^2-(\sqrt{3})^2} + \frac{(\sqrt{5}-\sqrt{3})^2}{(\sqrt{5})^2-(\sqrt{3})^2} \\
 &= \frac{(\sqrt{5}+\sqrt{3})^2 + (\sqrt{5}-\sqrt{3})^2}{5-3} \\
 &= \frac{2[(\sqrt{5})^2 + (\sqrt{3})^2]}{2} = 5+3=8.
 \end{aligned}$$

$$\begin{aligned}
 136. \text{ Given exp. } &= \frac{(\sqrt{2})^2}{(\sqrt{3})^2} \cdot \frac{(2+\sqrt{3})(2-\sqrt{3})}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{2}{3} \left[\frac{2^2 - (\sqrt{3})^2}{(\sqrt{3})^2 - 1^2} \right] \\
 &= \frac{2}{3} \left(\frac{4-3}{3-1} \right) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}.
 \end{aligned}$$

$$\begin{aligned}
 137. \text{ Given exp. } &= \frac{12}{3+(\sqrt{5}+2\sqrt{2})} \times \frac{3-(\sqrt{5}+2\sqrt{2})}{3-(\sqrt{5}+2\sqrt{2})} \\
 &= \frac{12(3-\sqrt{5}-2\sqrt{2})}{3^2-(\sqrt{5}+2\sqrt{2})^2} = \frac{12(3-\sqrt{5}-2\sqrt{2})}{9-(5+8+4\sqrt{10})} \\
 &= \frac{12(3-\sqrt{5}-2\sqrt{2})}{(-4-4\sqrt{10})} = \frac{3(\sqrt{5}+2\sqrt{2}-3)}{\sqrt{10}+1} \\
 &= \frac{3(\sqrt{5}+2\sqrt{2}-3)}{\sqrt{10}+1} \times \frac{\sqrt{10}-1}{\sqrt{10}-1}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{3\sqrt{50} - 3\sqrt{5} + 6\sqrt{20} - 6\sqrt{2} - 9\sqrt{10} + 9}{10 - 1} \\
 &= \frac{15\sqrt{2} - 3\sqrt{5} + 12\sqrt{5} - 6\sqrt{2} - 9\sqrt{10} + 9}{9} \\
 &= \frac{9\sqrt{2} + 9\sqrt{5} - 9\sqrt{10} + 9}{9} = 1 + \sqrt{2} + \sqrt{5} - \sqrt{10}.
 \end{aligned}$$

138. Given exp. = $\frac{1}{(\sqrt{2}-\sqrt{5})+\sqrt{3}} + \frac{1}{(\sqrt{2}-\sqrt{5})-\sqrt{3}}$

$$\begin{aligned}
 &= \frac{[(\sqrt{2}-\sqrt{5})-\sqrt{3}] + [(\sqrt{2}-\sqrt{5})+\sqrt{3}]}{[(\sqrt{2}-\sqrt{5})+\sqrt{3}][(\sqrt{2}-\sqrt{5})-\sqrt{3}]} \\
 &= \frac{2(\sqrt{2}-\sqrt{5})}{(\sqrt{2}-\sqrt{5})^2 - (\sqrt{3})^2} = \frac{2(\sqrt{2}-\sqrt{5})}{(2+5-2\sqrt{10})-3} \\
 &= \frac{2(\sqrt{2}-\sqrt{5})}{4-2\sqrt{10}} = \frac{\sqrt{2}-\sqrt{5}}{2-\sqrt{10}} \\
 &= \frac{\sqrt{2}-\sqrt{5}}{\sqrt{2}(\sqrt{2}-\sqrt{5})} = \frac{1}{\sqrt{2}}.
 \end{aligned}$$

139. $x + \frac{1}{x} = (7-4\sqrt{3}) + \frac{1}{(7-4\sqrt{3})} \times \frac{(7+4\sqrt{3})}{(7+4\sqrt{3})}$

$$\begin{aligned}
 &= (7-4\sqrt{3}) + \frac{(7+4\sqrt{3})}{(49-48)} \\
 &= (7-4\sqrt{3}) + (7+4\sqrt{3}) = 14.
 \end{aligned}$$

140. $x = 3 + \sqrt{8} \Rightarrow x^2 = (3 + \sqrt{8})^2 = 3^2 + (\sqrt{8})^2 + 2 \times 3 \times \sqrt{8}$

$$\begin{aligned}
 &= 9 + 8 + 6\sqrt{8} = 17 + 12\sqrt{2}.
 \end{aligned}$$

$$\begin{aligned}
 x^2 + \frac{1}{x^2} &= (17 + 12\sqrt{2}) + \frac{1}{(17 + 12\sqrt{2})} \times \frac{(17 - 12\sqrt{2})}{(17 - 12\sqrt{2})} \\
 &= (17 + 12\sqrt{2}) + \frac{(17 - 12\sqrt{2})}{289 - 288} \\
 &= (17 + 12\sqrt{2}) + (17 - 12\sqrt{2}) = 34.
 \end{aligned}$$

141. $a = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} \times \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}}$

$$\begin{aligned}
 &= \frac{(\sqrt{3} + \sqrt{2})^2}{(\sqrt{3})^2 - (\sqrt{2})^2} = \frac{3 + 2 + 2\sqrt{6}}{3 - 2} = 5 + 2\sqrt{6}.
 \end{aligned}$$

$$\begin{aligned}
 b &= \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}} = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}} \\
 &= \frac{(\sqrt{3} - \sqrt{2})^2}{(\sqrt{3})^2 - (\sqrt{2})^2} = \frac{3 + 2 - 2\sqrt{6}}{3 - 2} = 5 - 2\sqrt{6}.
 \end{aligned}$$

$$\begin{aligned}
 a^2 + b^2 &= (5 + 2\sqrt{6})^2 + (5 - 2\sqrt{6})^2 = 2[(5)^2 + (2\sqrt{6})^2] \\
 &= 2(25 + 24) = 2 \times 49 = 98.
 \end{aligned}$$

142. $a = \frac{(\sqrt{5}+1)}{(\sqrt{5}-1)} \times \frac{(\sqrt{5}+1)}{(\sqrt{5}+1)}$

$$\begin{aligned}
 &= \frac{(\sqrt{5}+1)^2}{(5-1)} = \frac{5+1+2\sqrt{5}}{4} = \left(\frac{3+\sqrt{5}}{2}\right).
 \end{aligned}$$

$$\begin{aligned}
 b &= \frac{(\sqrt{5}-1)}{(\sqrt{5}+1)} \times \frac{(\sqrt{5}-1)}{(\sqrt{5}-1)} = \frac{(\sqrt{5}-1)^2}{(5-1)} \\
 &= \frac{5+1-2\sqrt{5}}{4} = \left(\frac{3-\sqrt{5}}{2}\right).
 \end{aligned}$$

$$\begin{aligned}
 \therefore a^2 + b^2 &= \frac{(3+\sqrt{5})^2}{4} + \frac{(3-\sqrt{5})^2}{4} \\
 &= \frac{(3+\sqrt{5})^2 + (3-\sqrt{5})^2}{4} = \frac{2(9+5)}{4} = 7.
 \end{aligned}$$

Also, $ab = \frac{(3+\sqrt{5})}{2} \cdot \frac{(3-\sqrt{5})}{2} = \frac{(9-5)}{4} = 1.$

$$\therefore \frac{a^2 + ab + b^2}{a^2 - ab + b^2} = \frac{(a^2 + b^2) + ab}{(a^2 + b^2) - ab} = \frac{7+1}{7-1} = \frac{8}{6} = \frac{4}{3}.$$

143. $x = \sqrt{1 + \sqrt{1 + \sqrt{1 + \dots}}} \Leftrightarrow x = \sqrt{1+x}$

$$\begin{aligned}
 \Leftrightarrow x^2 &= 1 + x \Leftrightarrow x^2 - x - 1 = 0 \\
 \Leftrightarrow x &= \frac{1 \pm \sqrt{(-1)^2 - 4 \times 1 \times (-1)}}{2} \\
 \Leftrightarrow x &= \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}.
 \end{aligned}$$

Hence, positive value of x is $\frac{1+\sqrt{5}}{2}.$

144. Let $x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$

Then, $x = \sqrt{2+x} \Leftrightarrow x^2 = 2+x \Leftrightarrow x^2 - x - 2 = 0$

$$\begin{aligned}
 \Leftrightarrow x^2 - 2x + x - 2 &= 0 \\
 \Leftrightarrow x(x-2) + (x-2) &= 0 \Leftrightarrow (x-2)(x+1) = 0 \\
 \Leftrightarrow x &= 2. [\because x \neq -1]
 \end{aligned}$$

145. $a = \sqrt{3 + \sqrt{3 + \sqrt{3 + \dots}}} \Leftrightarrow a = \sqrt{3+a} \Leftrightarrow a^2 = 3+a$

$$\begin{aligned}
 \Leftrightarrow a^2 - a - 3 &= 0 \\
 \Leftrightarrow a &= \frac{1 \pm \sqrt{(-1)^2 - 4 \times 1 \times (-3)}}{2} \\
 \Leftrightarrow a &= \frac{1 \pm \sqrt{1+12}}{2} = \frac{1 \pm \sqrt{13}}{2} \Leftrightarrow a = \frac{1 + \sqrt{13}}{2} [\because a > 0] \\
 \Leftrightarrow a &= \frac{1 + 3.6}{2} = \frac{4.6}{2} = 2.3.
 \end{aligned}$$

$\therefore 2 < a < 3.$

146. $\frac{\sqrt{3+x} + \sqrt{3-x}}{\sqrt{3+x} - \sqrt{3-x}} = 2$

$$\begin{aligned}
 \Leftrightarrow \frac{\sqrt{3+x} + \sqrt{3-x}}{\sqrt{3+x} - \sqrt{3-x}} \times \frac{\sqrt{3+x} + \sqrt{3-x}}{\sqrt{3+x} + \sqrt{3-x}} &= 2 \\
 \Leftrightarrow \frac{(\sqrt{3+x} + \sqrt{3-x})^2}{(\sqrt{3+x})^2 - (\sqrt{3-x})^2} &= 2 \\
 \Leftrightarrow \frac{(3+x) + (3-x) + 2\sqrt{(3+x)(3-x)}}{(3+x) - (3-x)} &= 2 \\
 \Leftrightarrow 6 + 2\sqrt{9-x^2} = 4x \Leftrightarrow 2\sqrt{9-x^2} = 4x - 6
 \end{aligned}$$

$$162. \sqrt{.01} \times \sqrt[3]{.008} - .02 = \sqrt{(.1)^2} \times \sqrt[3]{(.2)^3} - .02 \\ = .1 \times .2 - .02 = .02 - .02 = 0.$$

$$163. \text{ Given exp. } = \sqrt[3]{\frac{0.008 + 0.000064}{0.064 + 0.000512}} = \sqrt[3]{\frac{0.008064}{0.064512}} \\ = \sqrt[3]{\frac{8064}{64512}} = \sqrt[3]{\frac{1}{8}} = \frac{1}{2} = 0.5.$$

$$164. \text{ Let } \sqrt[3]{3} = x.$$

$$\text{Then, } (\sqrt[3]{9} - \sqrt[3]{3} + 1) = (x^2 - x + 1) = \frac{x^3 + 1}{x + 1} = \frac{(\sqrt[3]{3})^3 + 1}{(\sqrt[3]{3} + 1)}.$$

$$\Rightarrow (\sqrt[3]{9} - \sqrt[3]{3} + 1)(\sqrt[3]{3} + 1) = (\sqrt[3]{3})^3 + 1$$

$$= 3 + 1 = 4, \text{ which is rational.}$$

$$165. \text{ Clearly, } (21)^3 = 9261 \text{ and } (22)^3 = 10648.$$

So, 9261 is the largest four-digit number which is a perfect cube.

$$166. 21600 = 2^5 \times 3^3 \times 5^2.$$

To make it a perfect cube, it must be multiplied by (2×5) , i.e., 10.

$$167. 3600 = 2^3 \times 5^2 \times 3^2 \times 2.$$

To make it a perfect cube, it must be divided by $5^2 \times 3^2 \times 2$, i.e. 450.

$$168. \text{ Required number to be added}$$

$$= 9^3 - 710 = 729 - 710 = 19.$$

$$169. \begin{array}{r|rr} 8 & 79 & 21 \\ 8 & 64 & \\ \hline 169 & 1521 & \\ 9 & 1521 & \\ & \times & \end{array}$$

$$\Rightarrow \sqrt{7921} = 89$$

$$170. \text{ Given } ? = \sqrt[4]{(625)^3}$$

$$= (625)^{\frac{3}{4}} = (5 \times 5 \times 5 \times 5)^{\frac{3}{4}}$$

$$= (5^4)^{\frac{3}{4}} = 5^3 = 125$$

$$171. \text{ Given } \sqrt{y} = 4x$$

$$\Rightarrow y = (4x)^2 = 16x^2$$

$$\Rightarrow y = 16x^2$$

$$\therefore \frac{x^2}{y} = \frac{1}{16}$$

$$172. \text{ Let the number be } x.$$

$$\text{Given: } \sqrt{2025.11} \times \sqrt{256.04} + \sqrt{399.95} \times \sqrt{?} = 33.98 \times 40$$

$$\sqrt{2025} \times \sqrt{256} + \sqrt{400} \times \sqrt{?} = 34 \times 40$$

$$45 \times 16 + 20 \times \sqrt{?} = 1360$$

$$20 \times \sqrt{?} = 1360 - 720 = 640$$

$$\sqrt{?} = \frac{640}{20} = 32$$

$$\Rightarrow ? = (32)^2$$

$$\therefore ? = 32 \times 32 = 1024$$

Hence, the number is 1024.

$$173. \frac{1}{\sqrt{28}} = \frac{1}{\sqrt{2 \times 2 \times 7}} = \frac{1}{2\sqrt{7}}$$

$$= \frac{\sqrt{7}}{2\sqrt{7} \times \sqrt{7}} = \frac{\sqrt{7}}{14}$$

$$= \frac{2.645}{14} = 0.189$$

$$174. = \left(\sqrt{\frac{25}{9}} - \sqrt{\frac{64}{81}} \right) + \sqrt{\frac{16}{324}}$$

$$= \left(\frac{5}{3} - \frac{8}{9} \right) + \frac{4}{18}$$

LCM of 3 and 9 is 9.

$$= \left(\frac{15-8}{9} \right) + \frac{2}{9} = \frac{7}{9} + \frac{2}{9}$$

$$= \frac{7}{9} + \frac{2}{9} = \frac{9}{9} = 1$$

$$175. \text{ Given: } 1728 + \sqrt[3]{262144} \times ? - 288 = 4491$$

$$\Rightarrow 1728 + \sqrt[3]{64 \times 64 \times 64} \times ?$$

$$= 4491 + 288$$

$$\Rightarrow \frac{1728}{64} \times ? = 4779$$

$$\Rightarrow 27 \times ? = 4779$$

$$\Rightarrow ? = \frac{4779}{27} = 177$$

$$176. \text{ Let the number be } a.$$

$$(\sqrt{7} + 11)^2$$

$$= a^{\frac{1}{3}} + 2\sqrt{847} + 122$$

$$\Rightarrow 7 + 121 + 22\sqrt{7}$$

$$= a^{\frac{1}{3}} + 22\sqrt{7} + 122$$

$$\Rightarrow 128 - 122 = a^{\frac{1}{3}}$$

$$\Rightarrow a^{\frac{1}{3}} = 6$$

$$\Rightarrow a = (6)^3 = 216$$

Hence, the number is 216.

$$177. \text{ Let the number be } x.$$

$$x - \sqrt{784} = 6 \times \sqrt{324}$$

$$x - 28 = 6 \times 18$$

$$x - 28 = 108$$

$$x = 108 + 28 = 136$$

$$178. \text{ Let the number be } x.$$

$$\text{Given } \sqrt{2116} - \sqrt{1600} = \sqrt{(x)}$$

$$46 - 40 = \sqrt{(x)}$$

$$\Rightarrow 6 = \sqrt{x}$$

$$\Rightarrow x = (6)^2 = 36$$

Hence, the number is 36.

179. Let the number be x .

$$\sqrt{(27+5x)+15} = 5.4+6+0.3$$

$$\Rightarrow \sqrt{\frac{27+5x}{15}} = \frac{5.4}{6} + 0.3$$

$$\Rightarrow \sqrt{\frac{27 \times x}{5 \times 15}} = \frac{5.4}{6} + 0.3$$

$$\Rightarrow \sqrt{\frac{9 \times x}{5 \times 5}} = 0.9 + 0.3$$

$$\Rightarrow \frac{3}{5} \sqrt{x} = 1.2$$

$$\Rightarrow \sqrt{x} = \frac{1.2 \times 5}{3} = 2 \Rightarrow x = (2)^2$$

$$\Rightarrow x = 2 \times 2 = 4^2$$

180. Given $\sqrt{24+0.5+1} + \sqrt{18+0.6+6}$

$$= \sqrt{24 \times \frac{1}{0.5} + 1} + \sqrt{18 \times \frac{1}{0.6} + 6}$$

$$= \sqrt{24 \times \frac{10}{5} + 1} + \sqrt{\frac{18 \times 10}{6} + 6}$$

$$= \sqrt{48+1} + \sqrt{30+6}$$

$$\sqrt{49} + \sqrt{36} = 7 + 6 = 13$$

181. Given $(\sqrt{63} + \sqrt{252}) \times (\sqrt{175} + \sqrt{28})$

$$= (\sqrt{63} + \sqrt{4 \times 63}) \times (\sqrt{25 \times 7} + \sqrt{4 \times 7})$$

$$= (\sqrt{7 \times 9} + \sqrt{4 \times 9 \times 7}) \times (\sqrt{5 \times 5 \times 7} + \sqrt{2 \times 2 \times 7})$$

$$= (3\sqrt{7} + 6\sqrt{7}) \times (5\sqrt{7} + 2\sqrt{7})$$

$$= 9\sqrt{7} \times 7\sqrt{7} = 441$$

182. Given $9x^2 + 25 - 30x$

$$\text{We have to find } \sqrt{9x^2 + 25 - 30x}$$

$$= \sqrt{(3x)^2 - 2 \cdot 3x \cdot 5 + (-5)^2} \quad \left\{ \because a^2 - 2ab + b^2 = (a-b)^2 \right\}$$

$$= \sqrt{(3x-5)^2} = 3x-5$$

183. $\sqrt{\frac{3}{11}} = \sqrt{\frac{3 \times 11}{11 \times 11}} = \frac{\sqrt{33}}{11}$

$$= \frac{5.745}{11} = 0.5223$$

184. Let the number be x .

$$\text{Given } \sqrt{x} + 14 = \sqrt{2601}$$

$$\text{or, } \sqrt{x} = 51 - 14 = 37$$

$$\text{or } x = 37^2 = 1369$$

185. $a = \frac{\sqrt{3}}{2}$ (given)

$$\therefore \sqrt{1+a} + \sqrt{1-a}$$

$$= \sqrt{1 + \frac{\sqrt{3}}{2}} + \sqrt{1 - \frac{\sqrt{3}}{2}}$$

$$= \sqrt{\frac{2+\sqrt{3}}{2}} + \sqrt{\frac{2-\sqrt{3}}{2}}$$

$$= \sqrt{\frac{2(2+\sqrt{3})}{4}} + \sqrt{\frac{2(2-\sqrt{3})}{4}}$$

$$= \sqrt{\frac{4+2\sqrt{3}}{4}} + \sqrt{\frac{4-2\sqrt{3}}{4}}$$

$$= \sqrt{\frac{3+1+2 \times \sqrt{3} \times 1}{2}} + \sqrt{\frac{3+1-2 \times \sqrt{3} \times 1}{2}}$$

$$\left\{ \begin{array}{l} \sqrt{3}^2 + (1)^2 - 2 \times \sqrt{3} \times 1 = (\sqrt{3}-1)^2 \\ (\sqrt{3})^2 + (1)^2 + 2 \times \sqrt{3} \times 1 = (\sqrt{3}+1)^2 \\ \therefore a^2 + b^2 + 2ab = (a+b)^2 \\ a^2 + b^2 - 2ab = (a-b)^2 \end{array} \right.$$

$$= \frac{\sqrt{(\sqrt{3}+1)^2}}{2} + \frac{\sqrt{(\sqrt{3}-1)^2}}{2}$$

$$= \frac{\sqrt{3}+1 + \sqrt{3}-1}{2}$$

$$= \frac{2\sqrt{3}}{2} = \sqrt{3}$$

186. Given $\frac{5+\sqrt{10}}{5\sqrt{5}-2\sqrt{20}-\sqrt{32}+\sqrt{50}}$

$$= \frac{5+\sqrt{10}}{5\sqrt{5}-2 \times 2\sqrt{5}-2 \times 2\sqrt{2}+5\sqrt{2}}$$

$$= \frac{5+\sqrt{10}}{5\sqrt{5}-4\sqrt{5}-4\sqrt{2}+5\sqrt{2}}$$

$$= \frac{5+\sqrt{10}}{\sqrt{5}+\sqrt{2}} = \frac{\sqrt{5}(\sqrt{5}+\sqrt{2})}{\sqrt{5}+\sqrt{2}} = \sqrt{5}$$

187. Given: $\frac{(0.75)^3}{1-0.75} + [0.75 + (0.75)^2 + 1]$

$$= \frac{(0.75)^2 \times 0.75}{0.25} + [0.75 + 0.5625 + 1]$$

$$= 0.5625 \times 3 + [0.75 + 0.5625 + 1]$$

$$= 1.6875 + 2.3125 = 4$$

$$\text{Square root of } 4 = 2$$

188. Given $\sqrt{10+2\sqrt{6}+2\sqrt{10}+2\sqrt{15}}$

$$= \sqrt{10+2 \times \sqrt{3} \times \sqrt{2} + 2 \times \sqrt{2} \times \sqrt{5} + 2 \times \sqrt{3} \times \sqrt{5}}$$

$$= \sqrt{2+3+5+2 \times \sqrt{2} \times \sqrt{3} + 2 \times \sqrt{5} \times \sqrt{2} + 2 \times \sqrt{5} \times \sqrt{3}}$$

$$= \sqrt{(\sqrt{2})^2 + (\sqrt{3})^2 + (\sqrt{5})^2 + 2 \times \sqrt{2} \times \sqrt{3} + 2 \times \sqrt{5} \times \sqrt{2} + 2 \times \sqrt{5} \times \sqrt{3}}$$

$$\left\{ \because a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = (a+b+c)^2 \right\}$$

$$= \sqrt{(\sqrt{2} + \sqrt{3} + \sqrt{5})^2}$$

$$= (\sqrt{2} + \sqrt{3} + \sqrt{5})$$